Submitted to the Annals of Statistics

EFFICIENT ESTIMATION IN SEMIPARAMETRIC ACCELERATED LIFE MODELS FOR RECURRENT EVENTS

BY EMMANUEL DJEGOU[1,a], AKIM ADEKPEDJOU[2,b]

[1]*Mathematics and Statistics, Missouri University of Science and Technology, Rolla, MO 65409, U.S.A.* , [a]*emdwmt@mst.edu*

[2]*Mathematics and Statistics, Missouri University of Science and Technology, Rolla, MO 65409, U.S.A.,* [b]*akima@mst.edu*

Accelerated life models, like the accelerated failure time (AFT) model, relate covariates to log event times under right-censoring and extend naturally to recurrent events via the accelerated gap time (AGT) model. While accelerated life models offer intuitive interpretations, serving as meaningful alternative to the Cox model, they often overlook the impact of interventions between events—common in fields like reliability engineering and biomedical research. We propose a broader class of semiparametric accelerated life models that incorporate an effective age process to account for such interventions. To address estimation challenges from the infinite-dimensional baseline hazard and non-monotonic score functions, we develop a sample-based weighted efficient score function using parametric submodels. The resulting estimators are consistent and asymptotically normal. Simulations and a real biomedical dataset demonstrate the method's effectiveness.

1. Introduction. In reliability engineering and medical research, failure time data are routinely collected for units or subjects, along with relevant covariates. For example, in reliability studies, factors such as material properties, operating conditions, maintenance history, and usage intensity affect component lifespans, while in medical research, variables including age, sex, blood pressure, cholesterol levels, and lifestyle habits influence patient survival. Regression models are therefore commonly used to characterize such relationships. The Cox (1972) model, which links covariates multiplicatively to the hazard function, is particularly popular. However, its proportional-hazards assumption prevents it from directly modeling how covariates accelerate or decelerate event times. In contrast, the AFT model assumes that covariates act multiplicatively on the event time, which makes it more attractive than the Cox model (Reid, 1994).

The AFT model, when expressed as a log-linear function of the covariates, is considered parametric (pAFT) if the distribution of the error term is specified and semiparametric (sAFT) if it is left unspecified. Specifically, for the sAFT model, Prentice (1978) introduced an estimation method for the regression coefficients based on the invertibility of weighted linear rank estimating equations. Tsiatis (1990) later established the asymptotic properties of this class of estimators. In parallel, Buckley and James (1979) introduced a least-squares approach that accommodates censored data to estimate the regression parameters, laying the foundation for inference under the sAFT model. Building on these results, Ritov (1990) examined Buckley–James-type estimating equations, demonstrating that they yield consistent and asymptotically normal estimators. In addition, he showed that these estimators attain the same efficiency as those derived from Prentice's linear rank-based method.

Recurrent event data, where subjects experience multiple event occurrences over time, arise frequently in longitudinal studies. A notable advancement in modeling such data is

MSC2020 subject classifications: Primary 00X00, 00X00; secondary 00X00.

Keywords and phrases: Accelerated life models, Effective age, Semiparametric Estimation, Recurrent events, Censoring, Time-to-failure data.

the semiparametric AGT (sAGT) model introduced by Strawderman (2005), which models the logarithm of the gap times between events as a linear function of covariates. This formulation captures how covariates stretch or compress the intervals between successive events. Inference for regression parameters is based on semiparametric efficiency theory, achieved through projection onto the tangent space of the baseline hazard, extending Prentice's weighted linear rank estimating equations to recurrent gap times. Consistency and asymptotic normality were established under Gehan (1965) and log-rank (Mantel, 1966) weights using empirical process and weak convergence theory.

While accelerated life models—such as the sAFT and sAGT frameworks—have been widely studied, extensions that accommodate greater modeling complexity remain relatively limited. In particular, the potential influence of interventions applied after each event occurrence on the timing of future events has not been fully explored. In reliability engineering, for example, when a component fails, corrective actions such as part replacement, recalibration, or operational adjustments are typically carried out. Systems, referring to either units or subjects depending on the context, that undergo effective maintenance tend to experience longer intervals before the next failure, while inadequate interventions may result in shorter times to recurrence. This impact is captured through an adjustment to the system's effective age, a concept used to model how maintenance alters the recurrence process.

Originally introduced by Kijima et al. (1988), the idea of effective age has been applied in various fields, including real estate (Robwes, 1977; Wil, 2008; Mil, 2018) to evaluate maintenance effects on depreciation, actuarial science (Latu, 2019) to model early retirement, and medicine (Gonzalez, 2005; Han, 2007; Pea, 2007) to track patient health dynamics. We extend this idea to accelerated life models, incorporating covariates and post-recurrence interventions under a log-linear effective age structure. A key advantage of the proposed model lies in its generality: it encompasses both the sAFT and sAGT models as special cases, corresponding to the *As Bad As Old* (ABAO) and *As Good As New* (AGAN) assumptions, respectively. Moreover, our approach offers a robust tool for analyzing censored recurrent event data, allowing for imperfect interventions through more flexible effective age processes.

The rest of the paper is organized as follows. Section 2 presents the accelerated effective age model and its formulation using counting processes. In Section 3, we extend this formulation to incorporate both calendar time and effective age time scales. Section 4 presents the proposed efficient estimation method. Section 5 illustrates the inference procedure using selected effective age processes based on existing models from the literature, while Section 6 establishes the corresponding asymptotic properties. Section 7 provides results from simulation studies that evaluate the finite-sample performance of the estimators. Section 8 illustrates the application of our model to a real-world dataset from rehospitalization studies. Finally, Section 9 concludes with a discussion of our findings and potential directions for future research. All proofs of the theoretical developments are provided in the Appendix.

2. Postulated Model. All random quantities are defined on a common probability space $(\Omega, \mathcal{F}, P)$. We consider a study in which the aging of n independent systems is of interest, with the possibility of imperfect interventions after each recurrence. Each system is observed over the interval $[0, \tau]$. The τ's are i.i.d. right censoring random variables whose unknown probability distribution function is $G(\cdot)$. Since the censorship mechanism in most observational study is unknown, we will assume that the censoring times are non-informative. Let $S_0 \equiv 0 < S_1 < S_2 < S_3 < \cdots$ denote the calendar times of event occurrences, with associated gap times $T_j = S_j - S_{j-1}$. Since the duration of each intervention is considered negligible and is not included in the analysis, we assume that the S_j's represent the times at which interventions occur, while T_j represents the time increment between the $(j-1)$-th and j-th interventions. A p-dimensional vector of time-fixed covariates, $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_p)'$, is also

recorded for each system. These covariates capture characteristics such as frequency of use, weather exposure, road conditions, and operating hours, as well as lifestyle habits and demographic information (such as sex and age) of patients, all of which are presumed to impact the system's aging process.

To introduce our model via a counting process formulation, we consider the following sequence of observable processes at time $s \in \mathcal{S}$: $\{(N_i^\dagger(s), Y_i^\dagger(s), \mathcal{E}_i(s), \mathbf{Z}_i),\, i = 1, \ldots, n\}$, where $\{Y_i^\dagger(s) = I(\tau_i \geq s) : s \in \mathcal{S}\}$ denotes the "at-risk" process, indicating whether the system is still at risk at calendar time s, and $\{\mathcal{E}_i(s) : s \in \mathcal{S}\}$ represents the effective age process, reflecting the system's age under accelerated consideration. Hence, conditional on the covariate vector $\mathbf{Z}$, the effective ages $\mathcal{E}_{i,0}, \mathcal{E}_{i,1}, \mathcal{E}_{i,2}, \ldots$ of the i-th system are observed at the j-th recurrence time $S_{i,j}$ before intervention, enabling inference to be made about the unobserved baseline effective ages. The observable effective age process $\mathcal{E}_i(s)$ is almost surely nonnegative, monotone, and differentiable with a strictly positive derivative on each interval $(S_{i,j-1}, S_{i,j}]$, with initial value $\mathcal{E}_i(0) = \mathcal{E}_{i0} \geq 0$. The baseline effective age process, that is, without accounting for covariate-induced acceleration, is denoted by $\{\tilde{\mathcal{E}}_i(s) : s \in \mathcal{S}\}$ and is related to the observable effective age process through

$$\mathcal{E}_i(s) = \tilde{\mathcal{E}}_i(s)\, e^{-\boldsymbol{\theta}_0' \mathbf{Z}_i}, \quad s \in \mathcal{S}, \tag{2.1}$$

where $\tilde{\mathcal{E}}_{i,0}, \tilde{\mathcal{E}}_{i,1}, \tilde{\mathcal{E}}_{i,3}, \ldots$ is the resulting sequence of baseline effective ages for the i-th system.

Model (2.1) suggests that the covariate vector $\mathbf{Z}_i$ directly influences the baseline effective age $\tilde{\mathcal{E}}_i(s)$, by either "accelerating" or "decelerating" it via an exponenciated weighted linear combination of the covariates. Hence, the covariates can compress or stretch the age of the system. We define $\mathscr{F}_{s^-} = \sigma\{N_i^\dagger(v), Y_i^\dagger(v), \mathcal{E}_i(v), \boldsymbol{Z}_i,\, i = 1, \ldots, n;\, 0 \leq v < s\}$ for $s \in \mathcal{S}$, as the full history of the observed processes just prior to calendar time s. The symbol $\sigma\{\cdot\}$ denotes the sigma-algebra generated by the processes and variables listed within the brackets, capturing the information available at time s. Additionally, the process $\mathbf{F} = \{\mathscr{F}_s, s \in \mathcal{S}\}$ is increasing and right-continuous, due to the right-continuity of the processes $N_i^\dagger(v), Y_i^\dagger(v)$, and $\mathcal{E}_i(v)$. The data consist of n independent and identically distributed realizations

$$\boldsymbol{\mathcal{O}}_i(\tau_i) = (K_i, T_{i,1}, \ldots, T_{i,K_i}, \tau_i - S_{i,K_i}, \boldsymbol{Z}_i, \{\mathcal{E}_i(s) : s \in \mathcal{S}\}),$$

for $i = 1, \ldots, n$, where $K_i = N_i^\dagger((s \wedge \tau_i)^-)$ is the number of uncensored failure times. Under this class of effective age processes, Lemma 2.1 presents a probabilistic model using the $\mathbf{F}$-compensator of of the counting process $\mathbf{N}^\dagger(s) = \{N_1^\dagger(s), \cdots, N_n^\dagger(s) : s \in \mathcal{S}\}$.

LEMMA 2.1. *We postulate that* $\mathbf{A}^\dagger(s, \boldsymbol{\theta}) = \{A_1^\dagger(s, \boldsymbol{\theta}), \cdots, A_n^\dagger(s, \boldsymbol{\theta}) : s \in \mathcal{S}\}$ *is the* $\mathbf{F}$-*compensator of the multivariate counting process* $\mathbf{N}^\dagger(s), s \in \mathcal{S}$, *with*

$$A_i^\dagger(s, \boldsymbol{\theta}) = \int_0^s Y_i^\dagger(v) \lambda_0\big(\tilde{\mathcal{E}}_i(v)\big) e^{\boldsymbol{\theta}' \boldsymbol{Z}_i} dv, \text{ for } i = 1, \ldots, n, \tag{2.2}$$

where $\lambda_0(\cdot)$ *denotes the unknown continuous baseline hazard function, which characterizes the distribution of the initial failure time, and* $\boldsymbol{\theta}$ *is a* p*-dimensional regression parameter. Then, the vector* $\mathbf{M}^\dagger(s)$ *consists of* $M_i^\dagger(s) = N_i^\dagger(s) - A_i^\dagger(s, \boldsymbol{\theta}_0)$ *for* $i = 1, \ldots, n$, *where each* $M_i^\dagger(s)$ *is an orthogonal, square-integrable, zero-mean martingale with respect to* $\mathcal{F}_s$.

The effective age framework provides a flexible and unifying approach for modeling systems subject to interventions. Semiparametric models based on effective age processes have been proposed by Adekpedjou and Stocker (2015, 2019) for recurrent events under both multiplicative and additive intensity functions. However, their models do not incorporate covariate effects on the effective age within the $\mathbf{F}$-compensator. This limitation is addressed in Lemma 2.1 under our proposed accelerated life model defined in (2.1). As for

the form of the effective age process, it depends on the nature of the intervention. For example, *minimal intervention* ($\mathcal{E}(s) = s$) retains the system in an ABAO state, while *perfect intervention* ($\mathcal{E}(s) = s - S_{N^\dagger(s^-)}$) resets it to an AGAN state. A more general formulation is given by Block et al. (1985): let $B_{i,1}, B_{i,2}, \ldots$ be a sequence of i.i.d. Bernoulli(p) random variables, where a success indicates a perfect intervention and p may vary with time. Define the number of perfect interventions by $m_i(s) = \sum_{j=1}^{N_i^\dagger(s^-)} B_{i,j}(s)$, and let the corresponding perfect intervention times be given by the random indices $0 \equiv \Omega_{i,0} < \Omega_{i,1} < \cdots$, where $\Omega_{i,k} = \min\{j > \Omega_{i,k-1} : B_{i,j} = 1\}$. The effective age process is then defined as $\mathcal{E}_i(s) = s - S_{i,\Omega_{i,m_i(s^-)}}$. This construction recovers minimal intervention when $p = 0$ and perfect intervention when $p = 1$. The Brown-Proschan model (1983) arises as a special case with constant success probability. Further generalizations include Kijima's Type I and Type II models (1989), as well as the ARA models proposed by Doyen and Gaudoin (2004), which offer greater flexibility for modeling imperfect intervention schemes. These frameworks have been further studied in works such as Lindqvist (1999) and Gonzalez et al. (2005).

3. Doubly-Indexed Processes. Estimating the model parameters and establishing their asymptotic properties are challenging for two main reasons. First, the baseline hazard $\lambda_0(\cdot)$ depends on the effective age process $\tilde{\mathcal{E}}_i(\cdot)$. Second, semiparametric estimation must rely on counting and at-risk processes associated with the sequence of random effective age variables $\tilde{\mathcal{E}}_{i,j}$. To address these issues, we adopt the doubly-indexed processes introduced by Peña (2001), which facilitate transitioning between calendar time and effective age in gap time modeling. Over the domain $\mathcal{T} = [0, t^*]$, where t^* is a fixed maximum effective age, we define $R_i(s,t) = I\{\tilde{\mathcal{E}}_i(s) \le t\}$, along with the associated processes:

$$N_i(s,t) = \int_0^s R_i(v,t)\, dN_i^\dagger(v), \quad A_i(s,t) = \int_0^s R_i(v,t)\, dA_i^\dagger(v, \boldsymbol{\theta}), \tag{3.1}$$

$$M_i(s,t) = \int_0^s R_i(v,t)\, dM_i^\dagger(v);$$

where, $A_i^\dagger(v, \boldsymbol{\theta})$ denotes the compensator of $N_i^\dagger(v)$, and $M_i^\dagger(v)$ is the associated martingale as defined in Lemma 2.1. Specifically, the doubly-indexed counting process $N_i(s,t)$ is a Lebesgue–Stieltjes integral, where $N_i^\dagger(s)$ experiences jumps at the observed event times $S_{i,1}, S_{i,2}, S_{i,3}, \ldots$. Consequently, it takes the form

$$N_i(s, t \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} I\big(\tilde{\mathcal{E}}_i(S_{i,j}) \le t\big), \tag{3.2}$$

which counts the number of events observed over calendar time for which the corresponding effective age does not exceed t. Following this, Proposition 3.1 presents a change-of-variable technique that transforms stochastic integrals from calendar time to effective age, removing the dependence of the baseline hazard $\lambda_0(\cdot)$ on $\tilde{\mathcal{E}}_i(\cdot)$ in (2.2). To enable this, we restrict the effective age process between successive events for the ith system by defining $\mathcal{E}_{i,j}(s) = \mathcal{E}_i(s) I(S_{i,j-1} < s \le S_{i,j})$ for s such that $Y_i^\dagger(s) > 0$. We assume that $\mathcal{E}_{i,j}(s)$ is invertible and differentiable on $(S_{i,j-1}, S_{i,j}]$, and define $I_{\mathcal{E}_{i,j}}(t; u, v) = I\left(\mathcal{E}_{i,j}(u^+) < t \le \mathcal{E}_{i,j}(v)\right)$ for any $u < v$. The remainder follows the approach of Peña et al. (2001) and is omitted.

PROPOSITION 3.1. *Let $h(\cdot)$ be a bounded integrable function. For $(s,t) \in \mathcal{S} \times \mathcal{T}$:*

$$\int_0^s h\left(\tilde{\mathcal{E}}_i(v)\right) \lambda_0\left(\tilde{\mathcal{E}}_i(v)\right) e^{\boldsymbol{\theta}' \boldsymbol{Z}_i} Y_i^\dagger(v) dv = \int_0^\infty h(r) \lambda_0(r) Y_i(s, r \mid \boldsymbol{\theta}) dr, \tag{3.3}$$

where for $i = 1, \ldots, n$,

$$Y_i(s, r \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} \frac{I_{\mathcal{E}_{i,j-1}}(re^{-\boldsymbol{\theta}'\boldsymbol{Z}_i}; S_{i,j-1}, S_{i,j})}{\mathcal{E}'_{i,j-1}(\mathcal{E}^{-1}_{i,j-1}(re^{-\boldsymbol{\theta}'\boldsymbol{Z}_i}))} + \frac{I_{\mathcal{E}_{i,K_i}}(re^{-\boldsymbol{\theta}'\boldsymbol{Z}_i}; S_{i,N_i^\dagger(s^-)}, s \wedge \tau_i)}{\mathcal{E}'_{i,K_i}(\mathcal{E}^{-1}_{i,K_i}(re^{-\boldsymbol{\theta}'\boldsymbol{Z}_i}))}. \tag{3.4}$$

Proposition 3.1 shows that $Y_i(s, t \mid \boldsymbol{\theta})$ is the generalized at-risk process corresponding to $N_i(s, t \mid \boldsymbol{\theta})$, and under the sAGT model of Strawderman (2005), it coincides with the gap time version of $Y_i^\dagger(s)$, where $\mathcal{E}_i(s) = s - S_{i,N_i^\dagger(s-)}$. The associated process $M_i(s, t \mid \boldsymbol{\theta})$ satisfies the Doob–Meyer decomposition $M_i(s, t \mid \boldsymbol{\theta}) = N_i(s, t \mid \boldsymbol{\theta}) - A_i(s, t \mid \boldsymbol{\theta})$, with $A_i(s, t \mid \boldsymbol{\theta}) = \int_0^t \lambda_0(w) Y_i(s, w \mid \boldsymbol{\theta})\, dw$, where $Y_i(s, w \mid \boldsymbol{\theta})$ is defined in (3.4). Stochastic integration theory can be used to prove that $\{M_i(s, t \mid \boldsymbol{\theta}_0) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$ is a square-integrable martingale for fixed t, with mean zero with respect to $\mathcal{F}_s$; however, $\{M_i(s, t \mid \boldsymbol{\theta}_0) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$ is not a martingale for fixed s, due to the lack of a suitable filtration for the random effective age, though it remains a zero-mean process.

We now present Lemma 3.1 that allows us to perform time-scale transformations when integrating with respect to the doubly-indexed process $\{M_i(s, t \mid \boldsymbol{\theta}) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$. The proof of Lemma 3.1 uses similar techniques as that of Proposition 3.1. It's therefore omitted.

LEMMA 3.1. *Let* $\mathbf{H}_i(s, t) = (H_{i1}(s, t), \ldots, H_{ip}(s, t))'$ *be vector-valued processes on* $(\Omega, \mathcal{F}, \mathbf{P})$ *with each* $H_{ij}(\cdot, \cdot)$ *being bounded, and* $(s, t) \in \mathcal{S} \times \mathcal{T}$, *for* $i = 1, 2, \ldots, n$. *For each* $t \in \mathcal{T}$, *we assume that the process* $\{\mathbf{H}_i(s, t) : s \in \mathcal{S}\}$ *is* $\mathbf{F}$*-predictable. Hence, we have:*

$$\int_0^s \mathbf{H}_i\left(s, \tilde{\mathcal{E}}_i(v)\right) M_i(dv, t) \equiv \int_0^t \mathbf{H}_i(s, r)\, M_i(s, dr \mid \boldsymbol{\theta}), \tag{3.5}$$

for $i = 1, 2, \ldots, n$ *and* $(s, t) \in \mathcal{S} \times \mathcal{T}$. *Moreover, Lemma 2.1 implies that:*

$$\mathbb{E} \int_0^t \mathbf{H}_i(s, r)\, M_i(s, dr \mid \boldsymbol{\theta}_0) = 0, \tag{3.6}$$

$$Var \int_0^t \mathbf{H}_i(s, r)\, M_i(s, dr \mid \boldsymbol{\theta}_0) = \int_0^t \mathbb{E}\left[\mathbf{H}_i(s, r)^{\otimes 2}\, Y_i(r \mid \boldsymbol{\theta}_0)\right] \lambda_0(r) dr, \tag{3.7}$$

where $\mathbb{E}[\cdot]$ *and Var*$[\cdot]$ *denote the expectation and variance operators, respectively.*

To simplify the derivations in the estimation section, we introduce the following aggregated processes.

$$N(s, t \mid \boldsymbol{\theta}) = \sum_{i=1}^n N_i(s, t \mid \boldsymbol{\theta}), \quad Y(s, t \mid \boldsymbol{\theta}) = \sum_{i=1}^n Y_i(s, t \mid \boldsymbol{\theta}),$$

$$A(s, t \mid \boldsymbol{\theta}) = \sum_{i=1}^n A_i(s, t \mid \boldsymbol{\theta}), \quad M(s, t \mid \boldsymbol{\theta}) = \sum_{i=1}^n M_i(s, t \mid \boldsymbol{\theta}).$$

4. Estimation Procedure. Censoring in failure time data restricts information on event times, complicating parameter estimation by introducing bias and reducing efficiency. The model involves a finite-dimensional regression parameter $\boldsymbol{\theta}_0$ and an infinite-dimensional baseline cumulative hazard function $\Lambda_0(\cdot)$, constituting a semiparametric framework. Treating Λ_0 as a nuisance parameter, we define the true parameter value as $\boldsymbol{\delta}_0 = (\boldsymbol{\theta}_0, \Lambda_0)$. Our method constructs an efficient score function for $\boldsymbol{\theta}_0$ by imposing orthogonality conditions, ensuring regular, asymptotically linear estimators and avoiding super-efficiency issues (Le Cam, 1951). Additionally, we propose an Aalen-type estimator for $\Lambda_0(\cdot)$.

4.1. *Maximum Likelihood Estimation.* First, we construct the likelihood function based on the realizations of the counting process $N_i^\dagger(s)$, following the point-process approach developed by Jacod and others (Andersen et al., 1993). Specifically, the likelihood over the interval $[0, s]$ is

$$L(\boldsymbol{\delta}, s) \propto \prod_{v\in[0,s]} \prod_{i=1}^{n} \Big\{ dA_i^\dagger(v, \boldsymbol{\theta}) \Big\}^{dN_i^\dagger(v)} \exp\Big\{ -A^\dagger(s, \boldsymbol{\theta}) \Big\}, \tag{4.1}$$

where $dA_i^\dagger(v, \boldsymbol{\theta}) = \lambda_i(\mathcal{E}_i(v), \boldsymbol{\theta})\, dv = Y_i^\dagger(v)\, \lambda_0\big(\tilde{\mathcal{E}}_i(v)\big)\, e^{\boldsymbol{\theta}'\boldsymbol{Z}_i}\, dv$, and $A^\dagger(s, \boldsymbol{\theta})$ denotes its aggregated version. $\prod\!\!\!\!\!\int$ is the product integral, a continuous analogue of $\prod$, also used to express the survivor function via the cumulative hazard (Kalbfleisch and Prentice, 2002). Taking the log, differentiating with respect to $\boldsymbol{\theta}$, and assuming interchangeability of sum and integral, yields the score function for $\boldsymbol{\theta}_0$ at $\boldsymbol{\delta} = \boldsymbol{\delta}_0$:

$$U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^\infty \boldsymbol{Z}_i\, \psi\big(\tilde{\mathcal{E}}_i(v)\big)\, dM_i^\dagger(v), \tag{4.2}$$

as $s \to \infty$, where $\psi(r) = 1 + r\{\dot{\lambda}_0(r)/\lambda_0(r)\}$, with $\dot{\lambda}_0(u)$ the derivative of $\lambda_0(u)$ and $M_i^\dagger(v)$ as defined in Lemma 2.1.

Next, to derive the score function for the infinite-dimensional parameter $\Lambda_0(\cdot)$, we consider a family of parametric submodels $(\boldsymbol{\theta}_0, \Lambda_0) \to (\boldsymbol{\theta}_0, \Lambda_\gamma)$, with parametrization:

$$\Lambda_\gamma(t) = \int_0^t \lambda_0(w) e^{\gamma\kappa(w)} du, \tag{4.3}$$

for $\gamma \in \mathbb{R}$ and $\kappa(\cdot)$ is some bounded element of a suitable Hilbert space $\mathbb{H}$ satisfying:

$$\mathbb{H} = \left\{ \kappa(\cdot) \in \mathbb{R} : \mathbb{E}\left[\int_0^\infty \kappa(r) dM_1(r \mid \boldsymbol{\theta}) \right] = 0, \int_0^\infty \kappa^2(r) \lambda_0(r) \mathbb{E}\left[Y_1(r \mid \boldsymbol{\theta}) \right] dr < \infty \right\},$$

The true model $(\boldsymbol{\theta}_0, \Lambda_0)$ corresponds to $\gamma = 0$. Under parametrization (4.3), let $U_{\Lambda_0}^{\kappa}(\boldsymbol{\delta})$ denote the score function for $\Lambda_0(\cdot)$, obtained by differentiating the log-likelihood with respect to γ and evaluating at $\gamma = 0$. The resulting score function at the true parameter $\boldsymbol{\delta}_0$ is:

$$U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^\infty \kappa\big(\tilde{\mathcal{E}}_i(v)\big) dM_i^\dagger(v). \tag{4.4}$$

Hence, a parametric nuisance tangent space for $\Lambda_0(\cdot)$ is given by:

$$\mathcal{G}_\gamma = \left\{ \sum_{i=1}^{n} \int_0^\infty \mathcal{K}\left(\tilde{\mathcal{E}}_i(v)\right) dM_i^\dagger(v) : \mathcal{K}(\cdot) \in \mathbb{R}^p \text{ is an arbitrary function} \right\}.$$

with

$$\sum_{i=1}^{n} \int_0^\infty \mathcal{K}\left(\tilde{\mathcal{E}}_i(v)\right) dM_i^\dagger(v) = \sum_{i=1}^{n} \int_0^\infty \mathbf{B}\kappa\left(\tilde{\mathcal{E}}_i(v)\right) dM_i^\dagger(v) = \mathbf{B} U_{\Lambda}^{\kappa}(\boldsymbol{\delta}), \tag{4.5}$$

where $\mathbf{B} \in \mathbb{R}^p$ is arbitrary. Formally, the semiparametric nuisance tangent space $\mathcal{G}_{\text{sem}}$ is defined as the mean-square closure of the tangent spaces $\mathcal{G}_\gamma$ induced by all possible submodels. Intuitively, (4.5) indicates that $\mathcal{G}_{\text{sem}}$ can be characterized by allowing $\kappa(\cdot)$ to vary over the infinite-dimensional space of score functions $\mathbb{H}$ (van der Vaart, 1998). Note that $\mathcal{G}_{\text{sem}}$ is a subspace of $\mathbb{H}$.

We then restrict the log-likelihood—also used by Peñá et al. (2001) in a gap time setting—to effective ages $\tilde{\mathcal{E}}_i(s) \leq t$ over the intervals $[0, s]$ and $[0, t]$, using the indicator $R_i(s, t)$, motivated by technical considerations. Applying Lemmas 2.1 and 3.1, the resulting score functions for $\boldsymbol{\theta}_0$ and $\Lambda_0(\cdot)$ are expressed as integrals over the effective age.

$$U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^t \boldsymbol{Z}_i \psi(w)\, dM_i(w \mid \boldsymbol{\theta}_0) \tag{4.6}$$

and

$$U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^t \kappa(w)\, dM_i(w \mid \boldsymbol{\theta}_0). \tag{4.7}$$

Finally, we derive the efficient score function for $\boldsymbol{\theta}_0$ given by:

$$U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0) = U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0),$$

where $U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0)$ is the orthogonal projection of $U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0)$ onto the semiparametric nuisance tangent space for $\Lambda_0(\cdot)$. Specifically, $\xi(\cdot) = (\xi_1(\cdot), \ldots, \xi_p(\cdot))'$ is the solution to the following orthogonality condition:

$$\mathbb{E}\left[\left\{U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0)\right\} U_{\Lambda}^{\kappa\prime}(\boldsymbol{\delta}_0)\right] = \mathbf{0}_p, \text{ for any function } \kappa(\cdot) \in \mathbb{R}. \tag{4.8}$$

Let $\boldsymbol{z}^{(j)}(r \mid \boldsymbol{\theta}) = \mathbb{E}\left[\boldsymbol{Z}_1^{\otimes j} Y_1(r \mid \boldsymbol{\theta})\right]$ for $j = 0, 1, 2$, where $\boldsymbol{x}^{\otimes k}$ denotes 1, a $p \times 1$ vector $\boldsymbol{x}$, and a matrix $\boldsymbol{x}\boldsymbol{x}'$ for $k = 0, 1, 2$, respectively. Proposition 4.1 then provides the efficient score function and the corresponding efficiency bound for estimating $\boldsymbol{\theta}_0$—the semiparametric counterpart of the Cramér-Rao lower bound.

PROPOSITION 4.1. *We define $\xi(r) = \psi(r)\bar{\boldsymbol{Z}}(r)$, where $\bar{\boldsymbol{Z}}(r) = \bar{\boldsymbol{Z}}(r \mid \boldsymbol{\theta}_0)$, and $\bar{\boldsymbol{Z}}(r \mid \boldsymbol{\theta}) = \boldsymbol{z}^{(1)}(r \mid \boldsymbol{\theta}) / \boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta})$. Additionally, we assume $\int_0^\infty \psi^2(w)\lambda_0(w)\, dw < \infty$ to ensure integrability. Hence, the efficient score function for $\boldsymbol{\theta}_0$ at $\boldsymbol{\delta} = \boldsymbol{\delta}_0$ is given by:*

$$U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^t \psi(w) \left\{\boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(w)\right\} dM_i(w \mid \boldsymbol{\theta}_0). \tag{4.9}$$

and its covariance matrix is:

$$\Sigma_{\boldsymbol{\theta}}^{\psi,\text{eff}} = \int_0^t \mathbb{E}\left[\left\{\boldsymbol{Z}_1 - \bar{\boldsymbol{Z}}(w)\right\}^{\otimes 2} Y_1(w \mid \boldsymbol{\theta}_0)\right] \psi^2(w)\lambda_0(w)\, dw. \tag{4.10}$$

The semiparametric information bound $\mathcal{B}(\boldsymbol{\theta}_0)$ for estimating $\boldsymbol{\theta}_0$ is given by the inverse of $\Sigma_{\boldsymbol{\theta}}^{\psi,\text{eff}}$, provided it is positive definite.

An efficient score function and the corresponding semiparametric efficiency bound for $\Lambda_0(\cdot)$ can also be derived, following van der Vaart's (1998) framework. The following proposition summarizes these results.

PROPOSITION 4.2. *Under the conditions of Proposition 4.1, and for $\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0) > 0$ for $w \in [0, t]$, the efficient score function for $\Lambda_0(\cdot)$ at $\boldsymbol{\delta} = \boldsymbol{\delta}_0$ is given by:*

$$U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^\infty \left[\frac{I\{w \leq t\}}{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)} - Q_\psi(w)\, \phi_i(w)\right] dM_i(w \mid \boldsymbol{\theta}_0), \tag{4.11}$$

where $\phi_i(w) = \boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(w)$*, and* $Q_\psi(w) = \Delta'_\psi(w)\,\mathcal{B}(\boldsymbol{\theta}_0)\,\psi(w)$*, with*

$$\Delta_\psi(t) = \int_0^t \frac{\boldsymbol{z^{(1)}}(w \mid \boldsymbol{\theta}_0)}{\boldsymbol{z^{(0)}}(w \mid \boldsymbol{\theta}_0)} \psi(w)\lambda_0(w)\,dw. \tag{4.12}$$

and the efficiency bound $\mathcal{B}(\Lambda_0)$ *for estimating* $\Lambda_0(\cdot)$ *at time* t *is the inverse of:*

$$\int_0^t \frac{\lambda_0(w)}{\boldsymbol{z^{(0)}}(w \mid \boldsymbol{\theta}_0)}\,dw + \Delta'_\psi(t)\mathcal{B}(\boldsymbol{\theta_0})\Delta_\psi(t). \tag{4.13}$$

4.2. *Estimation of* $\boldsymbol{\delta}_0$. To estimate $\boldsymbol{\theta}_0$ using the efficient score function (4.9), we first replace $\bar{\boldsymbol{Z}}(t \mid \boldsymbol{\theta})$ with its empirical estimate:

$$\bar{\mathbb{Z}}(t \mid \theta) = \frac{\mathbb{Z}^{(1)}(t \mid \boldsymbol{\theta})}{\mathbb{Z}^{(0)}(t \mid \boldsymbol{\theta})}, \text{ with } \mathbb{Z}^{(k)}(t \mid \boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^n \boldsymbol{Z}_i^{\otimes k} Y_i(t \mid \boldsymbol{\theta}), \quad k = 0, 1, 2. \tag{4.14}$$

As for $\psi(\cdot)$, it requires estimating the baseline hazard $\lambda_0(r) = f_0(r)/(1 - F_0(r))$ and its derivative $\dot{\lambda}_0(r)$. A common method, following Wand and Jones (1995), applies kernel smoothing to a modified empirical distribution to avoid division by zero, with $\dot{\lambda}_0(r)$ obtained by differentiating the estimator (Scott, 1992). This approach, however, is prone to bias, sensitive to bandwidth choice, and converges slowly. Kalbfleisch and Prentice (2002) similarly highlighted challenges in estimating asymptotic variance due to the hazard derivative in AFT models. Alternatively, Henrik (1983) proposed smoothing the Nelson-Aalen estimator, which Patil, Wells, and Marron (1994) endorsed for its martingale form, practical value, and visual clarity. Still, Jin et al. (2003) noted the limited reliability of nonparametric density estimation under censoring.

To address these challenges, Tsiatis (1990) and Ritov (1990) proposed an empirical weight function $\mathbb{W}(\cdot \mid \boldsymbol{\theta})$ as a substitute for the unknown $\psi(\cdot)$. By replacing $\psi(\cdot)$ with this function and using $\bar{\mathbb{Z}}(\cdot \mid \boldsymbol{\theta})$ in place of $\bar{\boldsymbol{Z}}(\cdot \mid \boldsymbol{\theta})$ in (4.9), a sample-based weighted efficient score function is obtained:

$$U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \int_0^t \mathbb{W}(w \mid \boldsymbol{\theta}) \left[\boldsymbol{Z}_i - \bar{\mathbb{Z}}(w \mid \boldsymbol{\theta})\right] dN_i(w \mid \boldsymbol{\theta}). \tag{4.15}$$

Letting $t \to \infty$, and using $N_i(s, t \mid \boldsymbol{\theta})$ expressed as a sum in (3.2), the score (4.15) becomes:

$$U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \mathbb{W}\left(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta}\right) \left[\boldsymbol{Z}_i - \frac{\sum_{l=1}^n \boldsymbol{Z}_l Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})}{\sum_{l=1}^n Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})}\right], \tag{4.16}$$

with:

$$Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta}) = \sum_{r=1}^{K_l} \frac{I_{\mathcal{E}_{l,r-1}}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}'\boldsymbol{Z}_l}; S_{l,r-1}, S_{l,r})}{\mathcal{E}'_{l,r-1}(\mathcal{E}^{-1}_{l,r-1}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}'\boldsymbol{Z}_l}))} + \frac{I_{\mathcal{E}_{l,K_l}}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}'\boldsymbol{Z}_l}; S_{l,K_l}, \tau_l)}{\mathcal{E}'_{l,K_l}(\mathcal{E}^{-1}_{l,K_l}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}'\boldsymbol{Z}_l}))}. \tag{4.17}$$

The score function in (4.16) is similar to the weighted rank estimating function for censored data derived by Kalbfleisch and Prentice (1973). Therefore, $\widehat{\boldsymbol{\theta}}$ can be identified as the zero-crossing or minimizer of $\left\|U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta})\right\|$, where $\|\cdot\|$ denotes the Euclidean norm, though non-monotonicity may lead to multiple solutions. Since a plug-in estimator cannot be directly obtained, numerical methods like grid search work well for small covariates, while simulated annealing (Lin and Geyer, 1992) is more efficient for high-dimensional covariates.

A common weight choice is $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv \mathbb{Z}^{(0)}(t \mid \boldsymbol{\theta})$, which corresponds to the Gehan (1965) weight. This choice offers the advantage of improved numerical stability and guarantees the consistent convergence of $\widehat{\boldsymbol{\theta}}$. We refer to the corresponding score function as $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$. Taking the limit as $t \to \infty$, equation (4.15) simplifies to:

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n} \int_0^\infty \mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta}) \left[\boldsymbol{Z}_i - \bar{\mathbb{Z}}(w \mid \boldsymbol{\theta})\right] dN_i(w \mid \boldsymbol{\theta})$$

$$= \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})(\boldsymbol{Z}_i - \boldsymbol{Z}_l), \tag{4.18}$$

where $Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})$ is the generalized at-risk process defined in (4.17). Adopting the notation $\tilde{\mathcal{E}}_{lr}^{-i}(\boldsymbol{\theta}) = \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) e^{-\boldsymbol{\theta}' \boldsymbol{Z}_i}$, where

$$\tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) = \begin{cases} \mathcal{E}_{l,r-1}(S_{l,r}) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \text{for } r \le K_l, \\ \mathcal{E}_{l,K_l}(\tau_l) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \text{for } r = K_l + 1. \end{cases},$$

and

$$\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) = \begin{cases} \mathcal{E}_{l,r-1}(S_{l,r-1}) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \text{for } r \le K_l, \\ \mathcal{E}_{l,K_l}(S_{l,K_l}) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \text{for } r = K_l + 1 \end{cases},$$

(4.18) can then be further expressed as:

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \sum_{r=1}^{K_l+1} \frac{I(\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) \le \tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \le \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}))}{\mathcal{E}'_{l,r-1}(\mathcal{E}_{l,r-1}^{-1}(\tilde{\mathcal{E}}_{ij}^{-l}(\boldsymbol{\theta})))} (\boldsymbol{Z}_i - \boldsymbol{Z}_l). \tag{4.19}$$

Other options include the Mantel (1966) weight, $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv 1$, yielding a log-rank-type function $U_{\boldsymbol{\theta}}^{\mathbb{L},\text{eff}}(\boldsymbol{\delta})$, and the Fleming-Harrington (1991) family $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv \widehat{F}_0^{\beta}(t)\{1 - \widehat{F}_0^{\beta}(t)\}^{\eta}$, where $\widehat{F}_0^{\beta}(t)$ serves as an estimator for $\tilde{F}_0(\cdot)$. While the log-rank score is optimal when $\lambda_0(r)$ is specified under a Weibull distribution, it may still lack monotonicity and behave poorly like other general weights. Using the log-rank weight and letting $t \to \infty$, equation (4.15) simplifies to:

$$U_{\boldsymbol{\theta}}^{\mathbb{L},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n} \int_0^\infty \left[\boldsymbol{Z}_i - \bar{\mathbb{Z}}(w \mid \boldsymbol{\theta})\right] dN_i(w \mid \boldsymbol{\theta})$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \frac{Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})}{\sum_{l=1}^{n} Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})} (\boldsymbol{Z}_i - \boldsymbol{Z}_l). \tag{4.20}$$

This can be rewritten as:

$$U_{\boldsymbol{\theta}}^{\mathbb{L},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} p_l(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \mid \boldsymbol{\theta}))(\boldsymbol{Z}_i - \boldsymbol{Z}_l), \tag{4.21}$$

with:

$$p_l(t \mid \boldsymbol{\theta}) = \frac{Y_l(t \mid \boldsymbol{\theta})}{\sum_{l=1}^{n} Y_l(t \mid \boldsymbol{\theta})} = \frac{Y_l(t \mid \boldsymbol{\theta})}{Y(t \mid \boldsymbol{\theta}_0)}, \quad t \ge 0. \tag{4.22}$$

The function $p_l(t \mid \boldsymbol{\theta})$ parallels Andersen and Gill (1982), with the exponential term weighting the effective age. To avoid definition problems, we require that $Y(t \mid \boldsymbol{\theta}_0) > 0$ via $J(t \mid \boldsymbol{\theta}_0)$.

Finally, given a consistent estimator $\widehat{\boldsymbol{\theta}}$, the baseline cumulative baseline hazard $\Lambda_0(t)$ can be estimated via a method-of-moments approach, yielding a Nelson-Aalen-type estimator:

$$\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}_0) = \int_0^t \frac{J(w \mid \boldsymbol{\theta}_0)}{Y(w \mid \boldsymbol{\theta}_0)} dN(w \mid \boldsymbol{\theta}_0), \tag{4.23}$$

where $J(w \mid \boldsymbol{\theta}_0) = I\{Y(w \mid \boldsymbol{\theta}_0) > 0\}$ ensures the estimator is well-defined. This form naturally arises by considering the restricted score function in (4.7) and imposing

$$\int_0^t \kappa(w)\{dN(w \mid \boldsymbol{\theta}_0) - Y(w \mid \boldsymbol{\theta}_0) d\Lambda_0(w)\} = 0, \quad \forall \kappa \in \mathbb{H},$$

which implies $d\Lambda_0(w) = dN(w \mid \boldsymbol{\theta}_0)/Y(w \mid \boldsymbol{\theta}_0)$, leading directly to (4.23). Thus, (4.23) may also be viewed as a maximum likelihood estimator of $\Lambda_0(t)$. However, using $\widehat{\boldsymbol{\theta}}$ in place of $\boldsymbol{\theta}_0$ can introduce additional variability, potentially increasing the variance of $\widehat{\Lambda}_0(t \mid \widehat{\boldsymbol{\theta}})$ and reducing its efficiency.

5. Some Special Cases. We illustrate the inference procedure using specific choices of the effective age process from the literature, including those underlying the sAFT and sAGT models. Additional semiparametric examples are reviewed in Beutner (2023).

EXAMPLE 5.1. *When $\mathcal{E}_i(s) = s$ (minimal intervention), we have $\mathcal{E}_i^{-1}(s) = s$ and $\mathcal{E}_i'(s) = 1$ and the efficient score function (4.19) becomes:*

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} (\boldsymbol{Z}_i - \boldsymbol{Z}_l) I(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\boldsymbol{Z}_i - \boldsymbol{Z}_l)),$$

This is the gradient of the convex objective function:

$$L_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \left\{\log \tau_l - \log S_{i,j} - \boldsymbol{\theta}'(\boldsymbol{Z}_i - \boldsymbol{Z}_l)\right\}^+, \tag{5.1}$$

with $a^+ = \max(a, 0)$, similar to the sAFT model of Lin et al. (1998). However, their approach does not assume independence of the baseline gap times or a common conditional distribution given $\boldsymbol{Z}$. The minimizers of $L_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta})$ and $\|U_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta})\|$ are asymptotically equivalent (Fygenson and Ritov, 1994). Alternatively, using the log-rank weight results in

$$U_{\boldsymbol{\theta}}^{\mathbb{L},\mathit{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \frac{I\left(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\boldsymbol{Z}_i - \boldsymbol{Z}_l)\right)}{\sum_{l=1}^{n} I\left(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\boldsymbol{Z}_i - \boldsymbol{Z}_l)\right)} (\boldsymbol{Z}_i - \boldsymbol{Z}_l), \tag{5.2}$$

and given $\widehat{\boldsymbol{\theta}}$, we estimate $\Lambda_0(t)$ using the formula:

$$\widehat{\Lambda}_0(t \mid \widehat{\boldsymbol{\theta}}) = \sum_{i=1}^{n} \sum_{j=1}^{K_i} \frac{I\left(\log t - \log S_{i,j} \geq \widehat{\boldsymbol{\theta}}' \boldsymbol{Z}_i\right)}{\sum_{l=1}^{n} I\left(\log \tau_l - \log S_{i,j} \geq \widehat{\boldsymbol{\theta}}'(\boldsymbol{Z}_i - \boldsymbol{Z}_l)\right)}.$$

EXAMPLE 5.2. *When $\mathcal{E}_i(s) = s - S_{i,N_i^\dagger(s-)}$ (perfect intervention), we have $\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) = 0$, and $\mathcal{E}_i'(s) = 1$. Hence, $U_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta})$ in (4.19) is fully specified as follows:*

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\mathit{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \sum_{r=1}^{K_l+1} (\boldsymbol{Z}_i - \boldsymbol{Z}_l) I\left(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \leq \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta})\right),$$

with:

$$\tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) = \begin{cases} T_{l,r} e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \textit{for } r \le K_l, \\ (\tau_l - S_{l,K_l}) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \textit{for } r = K_l + 1. \end{cases},$$

which is the gradient of the convex objective function:

$$L_{\boldsymbol{\theta}}^{\mathbb{G},\textit{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \sum_{r=1}^{K_l+1} \left\{ \log \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) - \log \tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \right\}^{+}, \tag{5.3}$$

similar to the sAGT model of Strawderman (2005). However, in both examples, a reparameterization $\boldsymbol{\theta} \to -\boldsymbol{\theta}$ is required (Jin et al., 2003). Under the log-rank weight, $U_{\boldsymbol{\theta}}^{\mathbb{L},\textit{eff}}(\boldsymbol{\delta})$ becomes:

$$U_{\boldsymbol{\theta}}^{\mathbb{L},\textit{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n} \sum_{j=1}^{K_i} \sum_{l=1}^{n} \frac{\sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \le \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta})\right)}{\sum_{l=1}^{n} \sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \le \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta})\right)} (\boldsymbol{Z}_i - \boldsymbol{Z}_l). \tag{5.4}$$

with:

$$\tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) = \begin{cases} T_{l,r} e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \textit{for } r \le K_l, \\ (\tau_l - S_{l,K_l}) e^{\boldsymbol{\theta}' \boldsymbol{Z}_l} & \textit{for } r = K_l + 1. \end{cases}$$

Given $\widehat{\boldsymbol{\theta}}$, the following formula can be used to estimate $\Lambda_0(t)$:

$$\widehat{\Lambda}_0(t \mid \widehat{\boldsymbol{\theta}}) = \sum_{i=1}^{n} \sum_{j=1}^{K_i} \frac{I\left(\tilde{\mathcal{E}}_{ij}(\widehat{\boldsymbol{\theta}}) \le t\right)}{\sum_{l=1}^{n} \sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\widehat{\boldsymbol{\theta}}) \le \tilde{\mathcal{E}}_{lr}(\widehat{\boldsymbol{\theta}})\right)}.$$

First, Examples 5.1, and 5.2 illustrate that a closed-form solution for $\widehat{\boldsymbol{\theta}}$ is unattainable, regardless of the chosen weight function. Moreover, the score function $U_{\boldsymbol{\theta}}^{\mathbb{L},\text{eff}}(\boldsymbol{\delta})$ is a weighted average of differences, where the presence of discontinuous indicator-based weights complicates the procedure. Second, while the Gehan estimating function $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ can be reformulated as a convex L_1-loss minimization problem, ensuring the existence of a solution, this does not readily extend to the log-rank or generalized weighted log-rank score functions. Hence, by estimating $\boldsymbol{\theta}$ using the Gehan weight, a consistent estimator can still be obtained for the log rank weight using linear programming techniques.

6. Asymptotic Results. In this section, we study the large sample properties [i.e $n \to \infty$] of the estimators $\widehat{\boldsymbol{\theta}}$ and $\widehat{\Lambda}_0(\cdot \mid \widehat{\boldsymbol{\theta}})$ at time t.

6.1. *Gehan-Weighted Estimation of $\boldsymbol{\theta}_0$.* Theorem 6.1 summarizes the large sample behavior of $\widehat{\boldsymbol{\theta}}_G$ under Conditions C1-C7.

THEOREM 6.1. *Let $\widehat{\boldsymbol{\theta}}_G = \arg\min_{\theta} \|U_{\boldsymbol{\theta}}^{\mathbb{G},\textit{eff}}(\boldsymbol{\delta})\|$ be the estimator of $\boldsymbol{\theta}_0$. Under suitable regularity conditions, $\widehat{\boldsymbol{\theta}}_G$ exists with probability converging to one as $n \to \infty$ and satisfies $n^{1/2}\left(\widehat{\boldsymbol{\theta}}_G - \boldsymbol{\theta}_0\right) \xrightarrow{d} N\left(0, \mathbb{A}_G\right)$, where $\mathbb{A}_G = \Phi_G^{-1\prime} \Sigma_{\boldsymbol{\theta}}^{\mathbb{G},\textit{eff}} \Phi_G^{-1}$ with*

$$\Sigma_{\boldsymbol{\theta}}^{\mathbb{G},\textit{eff}} = \int_0^{\infty} \left[\left(\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\right)^2 \boldsymbol{z}^{(2)}(w \mid \boldsymbol{\theta}_0) - \boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0) \left(\boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right)^{\otimes 2} \right] \lambda_0(w)\, dw,$$

and

$$\Phi_G = -\int_0^{\infty} \left[\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0) \boldsymbol{z}^{(2)}(w \mid \boldsymbol{\theta}_0) - \left(\boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right)^{\otimes 2} \right] \psi(w) \lambda_0(w)\, dw. \tag{6.1}$$

To establish these results, we rely on Theorems 4.1 and 4.2 from Ritov (1991). Theorem 4.1 demonstrates the asymptotic normality of $\widehat{\boldsymbol{\theta}}_G$, though it requires an $n^{1/2}$-consistent estimator—a condition that can be challenging to verify. Fortunately, the monotonicity of the Gehan-weighted score function $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ (as the gradient of a convex function $L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$) guarantees the existence of such an estimator and simplifies the proof of its $n^{1/2}$-consistency. In a subsequent section, we will show how $\mathbb{A}_G$ can be consistently estimated using a numerical approximation method.

6.2. *General Weighted Estimation of* $\boldsymbol{\theta}_0$. The monotonicity of $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ provides both computational and theoretical benefits that are lost with alternative weights like $\mathbb{W}(t \mid \boldsymbol{\theta}) = 1$, despite its potential optimality. Constructing an estimator based on $\widehat{\boldsymbol{\theta}}_G$ may improve the estimation results. Unlike general weight functions that require $\boldsymbol{\theta}_0$ to lie in a compact set for consistency, the proposed estimator in Theorem 6.2 obtained via linear approximation does not, since $\widehat{\boldsymbol{\theta}}_G$ is already $n^{1/2}$-consistent, thus simplifying computation. However, large-sample theory for general weights still demands stricter regularity conditions than those in Theorem 6.1. The one-step estimator's validity is ensured under Conditions C1–C10.

THEOREM 6.2. *Let $\widehat{\boldsymbol{\theta}}_W$ be defined as $\widehat{\boldsymbol{\theta}}_W = \widehat{\boldsymbol{\theta}}_G - n^{-1}\widehat{\Phi}_W^{-1} U_{\boldsymbol{\theta}}^{\mathbb{W},\textit{eff}}(\widehat{\boldsymbol{\delta}}_G)$, with $\widehat{\Phi}_W$ a consistent estimator of Φ_W. Then $\widehat{\boldsymbol{\theta}}_W$ exists with probability converging to one as $n \to \infty$ and $n^{1/2}\Big(\widehat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0\Big) \xrightarrow{d} N\Big(0, \mathbb{A}_W\Big)$, with $\mathbb{A}_W = \Phi_W^{-1\prime}\Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\textit{eff}}\Phi_W^{-1}$, where*

$$\Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\textit{eff}} = \int_0^{\bar{\tau}} w^2(r \mid \boldsymbol{\theta}_0)\, v(r \mid \boldsymbol{\theta}_0)\, \boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0)\, \lambda_0(r)\, dr,$$

$$\Phi_W = -\int_0^{\bar{\tau}} w(r \mid \boldsymbol{\theta}_0)\, v(r \mid \boldsymbol{\theta}_0)\, \boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0)\, \psi(r)\, \lambda_0(r)\, dr, \tag{6.2}$$

with $v(r \mid \boldsymbol{\theta}_0) = \boldsymbol{z}^{(2)}(r \mid \boldsymbol{\theta}_0)/\boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0) - \big\{\boldsymbol{z}^{(1)}(r \mid \boldsymbol{\theta}_0)/\boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0)\big\}^{\otimes 2}$; $\bar{\tau}$ and $w(r \mid \boldsymbol{\theta}_0)$ defined in Conditions C8 and C9, respectively.

To compute $\widehat{\boldsymbol{\theta}}_W$, a consistent estimator of Φ_W is required. However, this does not incur additional computational cost, as an estimate of Φ_W is already necessary for computing $\mathbb{A}_W$. The procedures for estimating $\widehat{\boldsymbol{\theta}}_W$ and its variance are presented in the next section.

6.3. *Estimating $\mathbb{A}_G$ and $\mathbb{A}_W$.* The main challenge in estimating the variances in Theorems 6.1 and 6.2 is accurately estimating Φ_G and Φ_W due to their dependence on $\psi(\cdot)$. To address this issue, we adopt the variance estimation method proposed by Huang (2002a) and later adapted by Strawderman (2005), which has been shown to yield reliable results. This technique, similar to numerical differentiation, is based on the asymptotic linearity of $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ in a neighborhood of $\boldsymbol{\theta}_0$. Under Conditions C1–C5, $\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}$ is consistent by Corollary C.1, and $\widehat{\mathbb{A}}_G$ consistently estimates $\mathbb{A}_G$, as established in Theorem 6.1.

ALGORITHM 1.

Phase 1: Compute $\widehat{\boldsymbol{\theta}}_G$ by minimizing $n^{-1}L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$, and estimate

$$\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}} = \int_0^{\infty}\Big[\mathbb{Z}^{(0)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big)\mathbb{Z}^{(2)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big) - \Big\{\mathbb{Z}^{(1)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big)\Big\}^{\otimes 2}\Big]\, d\mathbb{N}^{(0)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big).$$

This can be rewritten as $\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K_i}\varphi\Big(\mathcal{E}_{i,j}e^{\widehat{\boldsymbol{\theta}}_G'\boldsymbol{Z}_i}\Big)$, with

$$\varphi(w) = \mathbb{Z}^{(0)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big)\mathbb{Z}^{(2)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big) - \Big\{\mathbb{Z}^{(1)}\big(w \mid \widehat{\boldsymbol{\theta}}_G\big)\Big\}^{\otimes 2}.$$

Phase 2: Determine

$$\widetilde{\boldsymbol{\theta}}_G^k = \arg\min_{\boldsymbol{\theta}} \left\| n^{-1} L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) - h_k \right\|, \text{ for } k = 1, \ldots, p,$$

where h_k is the k-th column of $\left(n^{-1}\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}\right)^{1/2}$.

Phase 3: Calculate

$$\widehat{\mathbb{A}}_G = \left(\widehat{\mathbb{A}}_G^{1/2}\right)^2, \quad \text{with} \quad \widehat{\mathbb{A}}_G^{1/2} = n^{1/2}\left(\widetilde{\boldsymbol{\theta}}_G^1 - \widehat{\boldsymbol{\theta}}_G, \ldots, \widetilde{\boldsymbol{\theta}}_G^p - \widehat{\boldsymbol{\theta}}_G\right).$$

The main advantage of Algorithm 1 is that it eliminates the need to approximate Ψ_G. However, it involves minimizing a nonsmooth objective function. While this approach performs well with Gehan weights, extending it to general weight functions is challenging because the resulting estimating functions are non-monotone, complicating optimization and causing instability in the minimizers. To address this, Lyu, Tianmeng et al. (2018) proposed an induced smoothing technique that produces a smooth estimating function, allowing the application of standard numerical methods for estimating regression coefficients and variances in recurrent gap time data.

As for $\widehat{\boldsymbol{\theta}}_W$, it depends directly on the approximation of Φ_W. Hence, an alternative method that utilizes the asymptotic linearity of $U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta})$ by sampling from the distribution of $\widehat{\boldsymbol{\theta}}_G$ can be used instead. Indeed, for any $0 < \Gamma < \infty$ and $\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \leq \Gamma n^{-\frac{1}{2}}$, under the regularity conditions specified in Theorem 6.2, the following Taylor series expansion holds:

$$n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}) - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0) = \Phi_W\left(\boldsymbol{\theta} - \boldsymbol{\theta}_0\right) + o_P\left(n^{-\frac{1}{2}}\right). \tag{6.3}$$

Thus, we randomly sample $\widetilde{\boldsymbol{\theta}}_k \sim N\left(\widehat{\boldsymbol{\theta}}_G, n^{-1}\widehat{\mathbb{A}}_G\right)$ for $k = 1, \ldots, B$ and applying equation (6.3) twice, we obtain:

$$n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\left(\widetilde{\boldsymbol{\delta}}_k\right) - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\left(\widehat{\boldsymbol{\delta}}_G\right) = \Phi_W\left(\widetilde{\boldsymbol{\theta}}_k - \widehat{\boldsymbol{\theta}}_G\right) + \epsilon_k \quad (k = 1, \ldots, B).$$

This implies that Φ_W can be viewed as the slope of a linear regression model with response $n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\left(\widetilde{\boldsymbol{\delta}}_k\right) - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\left(\widehat{\boldsymbol{\delta}}_G\right)$ and regressors $\widetilde{\boldsymbol{\theta}}_k - \widehat{\boldsymbol{\theta}}_G$ for $k = 1, \ldots, B$. Hence, Φ_W can be estimated as

$$\widetilde{\Phi}_{W,B} = \left[\left(\mathfrak{P}'\mathfrak{P}\right)^{-1}\mathfrak{P}'\mathfrak{Q}\right]^{(-1)},$$

where $A^{(-1)}$ represents matrix A without its first row and

$$\mathfrak{P} = \left[\mathbf{1}_B\left(\widetilde{\boldsymbol{\Theta}} - \widehat{\boldsymbol{\theta}}_G\mathbf{1}_B'\right)'\right], \quad \mathfrak{Q} = \left(n^{-1}\mathbf{U} - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\widehat{\boldsymbol{\delta}}_G)\mathbf{1}_B'\right)',$$

where $\mathbf{1}_B$ is a $B \times 1$ vector of ones,

$$\widetilde{\boldsymbol{\Theta}} = \left[\widetilde{\boldsymbol{\theta}}_1, \widetilde{\boldsymbol{\theta}}_2, \ldots, \widetilde{\boldsymbol{\theta}}_B\right], \text{ and } \mathbf{U} = \left[U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\widetilde{\boldsymbol{\delta}}_1), \ldots, U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\widetilde{\boldsymbol{\delta}}_B)\right].$$

As a result, we have the following theorem, derived under Conditions C1–C10:

THEOREM 6.3. *Let $\widehat{\Phi}_W = \lim_{B\to\infty}\widetilde{\Phi}_{W,B}$. Then $\widehat{\Phi}_W \to \Phi_W$ in probability and $\widehat{\mathbb{A}}_W = \widehat{\Phi}_W^{-1'}\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\widehat{\Phi}_W^{-1}$ is a consistent estimator of $\mathbb{A}_W$, with $\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}$ given by*

$$\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}} = \int_0^{\bar{\tau}} \mathbb{W}^2\left(w \mid \widehat{\boldsymbol{\theta}}_W\right)\left[\frac{\mathbb{Z}^{(2)}\left(w \mid \widehat{\boldsymbol{\theta}}_W\right)}{\mathbb{Z}^{(0)}\left(w \mid \widehat{\boldsymbol{\theta}}_W\right)} - \left\{\frac{\mathbb{Z}^{(1)}\left(w \mid \widehat{\boldsymbol{\theta}}_W\right)}{\mathbb{Z}^{(0)}\left(w \mid \widehat{\boldsymbol{\theta}}_W\right)}\right\}^{\otimes 2}\right]\mathbb{N}^{(0)}\left(dw \mid \widehat{\boldsymbol{\theta}}_W\right).$$

Since $\widetilde{\Phi}_{W,B}$ is not necessarily symmetric, we use the symmetric approximation

$$\widehat{\Phi}_{W,B} = \frac{1}{2}\left(\widetilde{\Phi}_{W,B} + \widetilde{\Phi}'_{W,B}\right).$$

The computational algorithm for estimating $\widehat{\boldsymbol{\theta}}_W$ and $\widehat{\mathbb{A}}_W$ is then summarized as follows.

ALGORITHM 2.

Phase 1: Apply Algorithm 1 to determine $\widehat{\boldsymbol{\theta}}_G$ and obtain an estimate for $\widehat{\mathbb{A}}_G$.
Phase 2: For a sufficiently large B, compute $\widehat{\Phi}_{W,B}$, then estimate $\widehat{\boldsymbol{\theta}}_W$ using:

$$\widehat{\boldsymbol{\theta}}_W = \widehat{\boldsymbol{\theta}}_G - n^{-1}\widehat{\Phi}_{W,B}^{-1} U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\left(\widehat{\boldsymbol{\delta}}_G\right).$$

Phase 3: Using $\widehat{\Phi}_{W,B}$ as a substitute for $\widehat{\Phi}_W$, compute $\widehat{\mathbb{A}}_W$ following Theorem 6.3.

The primary drawback of this algorithm is the increased computational cost required to evaluate $n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\widetilde{\boldsymbol{\delta}}_k)$ for each $k = 1, \ldots, B$. For comparison, a modified version of Algorithm 1 using the log-rank weight will be used in the simulation to approximate $\mathbb{A}_{\text{LR}}$.

6.4. *Estimation of* $\Lambda_0(\cdot)$. Assuming $\widehat{\boldsymbol{\theta}}_W$ is obtained via Theorem 6.2, $\Lambda_0(t)$ is estimated by $\widehat{\Lambda}_0(t \mid \widehat{\boldsymbol{\theta}}_W)$. In contrast, when using $\widehat{\boldsymbol{\theta}}_G$ derived from Theorem 6.1, use $\widehat{\Lambda}_0(t \mid \widehat{\boldsymbol{\theta}}_G)$ intead and Conditions C9–C10 are no longer required.

THEOREM 6.4. *Given $\widehat{\boldsymbol{\theta}}_W$, the process $n^{\frac{1}{2}}\left\{\widehat{\Lambda}_0(\cdot \mid \widehat{\boldsymbol{\theta}}_W) - \Lambda_0(\cdot)\right\}$ converges in distribution to a Gaussian process on $[0, \bar{\tau}]$ with the covariance function*

$$\mathbb{B}_{\Lambda_0}(r,t) = \int_0^{r\wedge t} \frac{\lambda_0(w)}{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)} dw + \Delta'_\psi(r)\mathbb{A}_W\Delta_\psi(t), \tag{6.4}$$

where $\Delta_\psi(t)$ is defined in (4.12). Furthermore, for any finite $\Gamma > 0$, we have

$$\sup_{\substack{t\in[0,\bar{\tau}] \\ n^{\frac{1}{2}}\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|\le\Gamma}} n^{\frac{1}{2}}\left|\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}) - \widehat{\Lambda}_0(t \mid \boldsymbol{\theta}_0) + \Delta'_\psi(t)(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\right| \xrightarrow{P} 0. \tag{6.5}$$

Setting $r = t$, a consistent variance estimator $\widehat{\mathbb{B}}_{\Lambda_0}(t)$ is obtained using $\widehat{\mathbb{A}}_W$ from Theorem 6.3 and a consistent estimator $\widehat{\Delta}(t)$ for $\Delta_\psi(t)$.

The linear approximation in equation (6.5) suggests that $\Delta_\psi(t)$ can be estimated in a manner analogous to Theorem 6.3. Following Andersen et al. (1993), a confidence interval $100(1-\alpha)\%$ for $\Lambda_0(t)$ can then be obtained using the following:

$$\widehat{\Lambda}_0\left(t \mid \widehat{\boldsymbol{\theta}}_W\right) \exp\left[\pm\frac{z_{\alpha/2}\widehat{\mathbb{B}}_{\Lambda_0}(t)}{n^{\frac{1}{2}}\widehat{\Lambda}_0\left(t \mid \widehat{\boldsymbol{\theta}}_W\right)}\right]$$

7. Simulation Studies. We assess the performance of the estimators $\widehat{\boldsymbol{\theta}}_G$ and $\widehat{\boldsymbol{\theta}}_{LR}$ under the weight functions of Gehan and log rank. The goal is to investigate the robustness of our proposed estimators in the sAFT and sAGT models. Although these models have previously been studied, they represent special cases of our proposed class. Their inclusion enables us to validate the performance of our general estimation procedure in well-understood settings.

In each scenario, we simulate recurrent event data for n independent systems, each characterized by a single covariate Z generated independently. Event times are generated through model (2.1). The effective age process $\mathcal{E}(s)$ is defined differently across experiments to reflect varying assumptions about how prior events and interventions affect the system's age. Each system is administratively censored at a time τ drawn independently from a $\mathrm{Uni}(0, 3.5)$ distribution, yielding approximately 20% censoring.

We repeat each experiment over 1000 replications for sample sizes $n \in \{50, 100\}$. The estimators $\widehat{\boldsymbol{\theta}}_G$ and $\widehat{\mathbb{A}}_G$ were computed using Algorithm 1, while Algorithm 2 was employed to obtain $\widehat{\boldsymbol{\theta}}_{\mathrm{LR}}$ and $\widehat{\mathbb{A}}_{\mathrm{LR}}$. The estimators are evaluated in terms of empirical bias, standard deviation, and the coverage probability of the 95% confidence intervals. The average estimated standard errors for $\widehat{\boldsymbol{\theta}}_G$ and $\widehat{\boldsymbol{\theta}}_{\mathrm{LR}}$, obtained using Huang's approximation technique, are reported under the column $\mathbb{E}[\mathrm{Est.\,SE}]$ (Huang). For the log-rank weight, the corresponding values under $\mathbb{E}[\mathrm{Est.\,SE}]$ (Reg) refer to the approximations based on Theorem 6.3. These estimates were not computed for the Gehan weight. Estimation of the standard errors was based on $B_1 = 100$ simulated values for $\widehat{\boldsymbol{\theta}}_{\mathrm{LR}}$ and $B_2 = 250$ for $\widehat{\Lambda}(\cdot \mid \widehat{\boldsymbol{\theta}}_G)$ and $\widehat{\Lambda}(\cdot \mid \widehat{\boldsymbol{\theta}}_{\mathrm{LR}})$.

Experiment 1. In the first experiment, we simulate data under the sAFT model, corresponding to an effective age process defined as $\mathcal{E}_i(s) = s$, i.e., calendar time. This setup assumes that prior events or interventions do not alter the effective age of the system, reflecting scenarios in which failures accumulate continuously over time without resets. Censoring is applied as described in the general setup. For each individual, the first gap time is drawn from an exponential distribution with rate $\lambda = 1.5$, and subsequent gap times are generated using Gumbel's (1960) bivariate exponential distribution with a correlation of 0.25 between successive gaps, which introduces positive temporal dependence between events. The observed event times are then computed as cumulative sums of the gap times, scaled by the factor $e^{-0.8Z_i}$, where $Z_i \sim \mathrm{Ber}(0.5)$. The cumulative hazard is evaluated at $s = 1$ and $s = 3$ as $\Lambda_0(1) = 1.18$ and $\Lambda_0(3) = 3.35$, respectively. This experiment evaluates the estimator's performance under positively correlated recurrent event processes, with always minimal intervention applied.

Experiment 2. In the second experiment, we consider the sAGT model, where the effective age resets after each event. Specifically, we define the effective age process as $\mathcal{E}_i(s) = s - S_{i,N_i(s^-)}$, reflecting the assumption that each recurrence initiates a new life cycle, independent of the past. Event times are computed as cumulative sums of the gap times, beginning from zero. The gap times are generated using the relationship $T_{i,j} \sim \tilde{T}_{i,j} e^{-0.8Z_i}$. The baseline hazard follows a Weibull distribution, resulting in a Power Law Process (PLP) for the initial intensity, given by $\lambda_0(\epsilon) = \alpha\gamma\epsilon^{\gamma-1}$ and cumulative hazard $\Lambda_0(\epsilon) = \alpha\epsilon^{\gamma}$, with parameters $\alpha = \gamma = 1.5$. $\Lambda_0(\epsilon)$ is evaluated at $\epsilon = 1$ and $\epsilon = 3$ as $\Lambda_0(1) = 1.5$ and $\Lambda_0(3) \approx 7.79$, respectively. $Z_i \sim \mathcal{N}(0, 1)$, and the censoring times τ are drawn, as described in the general setup. This simulation enables us to evaluate the performance of the estimation procedure when covariate effects act on the time between events rather than on calendar time.

Analysis Results. The simulations are conducted in MATLAB and the results are summarized in Table 1. According to Lin et al. (1998), the estimators $\widehat{\boldsymbol{\theta}}_G$ and $\widehat{\boldsymbol{\theta}}_{\mathrm{LR}}$ yield similar results when the model is correctly specified, since both are consistent. Our proposed methods demonstrate strong performance with moderate sample sizes, exhibiting minimal bias and 95% confidence interval coverage that closely matches theoretical expectations. Moreover, results remain stable across both Gehan and log-rank weights. Variance estimates obtained via Huang's algorithm and through the asymptotic linearity of the log-rank score function show comparable accuracy, with further improvement possible by increasing the resampling

iterations. These findings highlight the flexibility and effectiveness of accelerated life models in handling recurrent event data under different intervention schemes.

	Models							
	sAFT-Gumbel				sAGT-PLP			
	Log-rank		Gehan		Log-rank		Gehan	
Sample size (n)	50	100	50	100	50	100	50	100
$\lvert\,\mathrm{Bias}(\widehat{\boldsymbol{\theta}})\rvert$	0.28	0.29	0.16	0.17	0.22	0.23	0.13	0.14
$\mathrm{SE}(\widehat{\boldsymbol{\theta}})$	0.13	0.11	0.10	0.07	0.16	0.14	0.11	0.08
$\mathbb{E}[\mathrm{Est.\,SE}]$ (Huang)	0.79	0.83	0.44	0.46	0.82	0.83	0.33	0.34
$\mathbb{E}[\mathrm{Est.\,SE}]$ (Reg)	0.97	1.01	-	-	0.56	0.45	-	-
Coverage (95%) CI's (Wald Method)	0.99	0.99	0.94	0.93	0.92	0.92	0.70	0.77
Coverage (95%) CI's (Percentile Method)	0.93	0.97	0.96	0.92	0.96	0.96	0.96	0.96
$\lvert\,\mathrm{Bias}\{\widehat{\Lambda}(0.5)\}\rvert$	0.19	0.18	0.23	0.26	0.18	0.18	0.22	0.25
$\mathrm{SE}\left\{\widehat{\Lambda}_0(0.5)\right\}$	0.10	0.08	0.21	0.18	0.11	0.08	0.22	0.25
$\mathbb{E}[\mathrm{Est.\,SE}]$(Reg)	0.69	0.70	1.14	1.15	0.66	0.66	1.13	1.14
Coverage (95%) CI's (Wald Method)	0.97	0.99	0.94	0.93	0.97	0.97	0.92	0.92
Coverage (95%) CI's (Percentile Method)	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96
$\lvert\,\mathrm{Bias}\{\widehat{\Lambda}(1.5)\}\rvert$	0.54	0.54	0.99	0.96	0.48	0.49	1.01	0.96
$\mathrm{SE}\left\{\widehat{\Lambda}_0(1.5)\right\}$	0.26	0.19	0.41	0.30	0.25	0.21	0.42	0.31
$\mathbb{E}[\mathrm{Est.\,SE}]$(Reg)	1.75	1.78	2.21	2.18	1.61	1.63	2.20	2.17
Coverage (95%) CI's (Wald Method)	0.99	0.99	0.94	0.94	0.98	0.95	0.93	0.92
Coverage (95%) CI's (Percentile Method)	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96

TABLE 1

Simulation Results for sAFT-Gumbel and sAGT-PLP Models.

8. Data Analysis. Colorectal cancer, a common malignancy in both men and women, typically originates from polyps in the colon or rectum, with risk factors including age, genetics, lifestyle, and conditions such as inflammatory bowel disease. Treatment generally begins with surgery, followed by chemotherapy or radiation; however, postoperative readmissions are frequent due to complications, recurrence, or treatment side effects. In this section, we apply our model to a dataset on hospital readmissions in colorectal cancer patients following surgery. The dataset includes 403 patients, divided into two groups based on chemotherapy status. The data come from a study by Gonzalez et al. (2005), published in the *Journal of Epidemiology and Community Health* and available in the R package `frailtypack`, which examines sex differences in readmission rates. We fit the accelerated life model in (2.1) to capture how patients age over time, incorporating treatment history and comorbid conditions, enabling a clearer understanding of readmission risk and supporting more informed clinical decisions. To evaluate the effectiveness of chemotherapy, we assess patient condition following readmission using three time-fixed covariates: chemotherapy status, Dukes' stage, and sex. Chemotherapy is coded as a binary variable (chemo $= 1$ for treated, chemo $= 0$ for untreated). Dukes' stage is categorized as 1 for stages A–B, 2 for stage C, and 3 for stage D, with lower values indicating less advanced cancer. Sex is coded as 0 for male and 1 for female. As noted by Stocker and Adekpedjou (2020), model identifiability relies on directly observing the effective age process over time, and its absence adds complexity to the modeling task. For illustration, we consider two effective age processes—always minimal and always perfect interventions—corresponding to the sAFT-TIC and sAGT-TIC models, where TIC means the covariates are constant over time. Table 1 presents the parameter estimates, standard errors, test statistics, and p-values for both models. The estimates using the Log-rank weight are computed through Algorithm 2, employing $B_1 = 2000$ resampling iterations.

		Log-rank Weight				Gehan Weight			
Model	**Covariate**	$\widehat{\theta}_{LR}$	$\widehat{\mathbb{A}}_{LR}$	95% CI	p-val	$\widehat{\theta}_{G}$	$\widehat{\mathbb{A}}_{G}$	95% CI	p-val
sAFT-TIC	Chemo	0.36	2.00	(-2.53, 3.26)	0.80	0.33	2.40	(-2.83, 3.50)	0.83
	Sex	-0.14	2.11	(-3.12, 2.83)	0.92	-0.15	4.13	(-3.44, 3.12)	0.92
	Dukes	0.25	1.21	(-2.00, 2.51)	0.81	0.28	3.74	(-3.67, 4.23)	0.88
sAGT-TIC	Chemo	0.36	2.00	(-2.53, 3.26)	0.80	0.33	2.40	(-2.83, 3.50)	0.83
	Sex	-0.14	2.11	(-3.12, 2.83)	0.92	-0.15	4.13	(-3.44, 3.12)	0.92
	Dukes	0.25	1.21	(-2.00, 2.51)	0.81	0.28	3.74	(-3.67, 4.23)	0.88

TABLE 2

Comparison of parameter estimates, variances, 95% confidence intervals, and p-values for the sAFT-TIC and sAGT-TDC models under Log-rank and Gehan weights.

Figure 1 illustrates the differences in baseline survival function estimates between the two models.

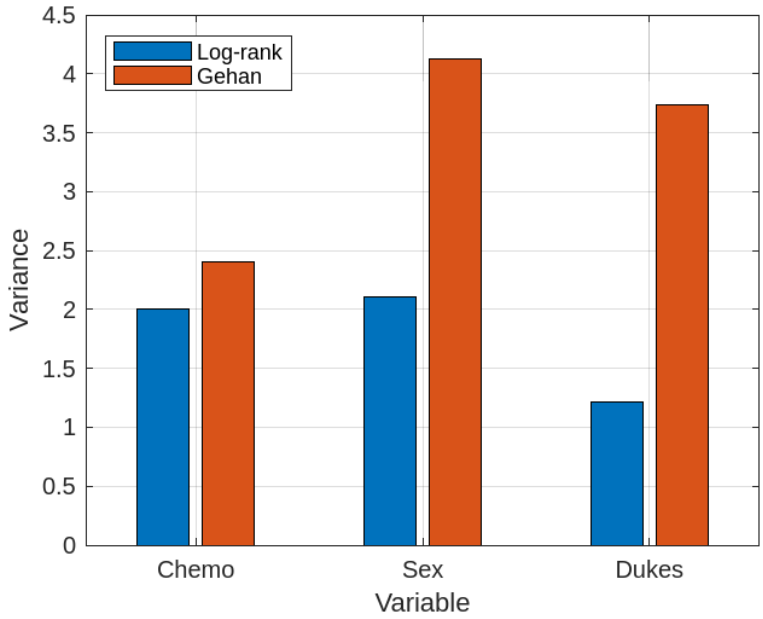

Fig 1: Bar Chart Comparing Variances for Log-rank and Gehan Weights.

9. Discussion. Future work could address the model's non-smoothness by applying induced smoothing techniques (Lyu et al., 2018), enabling smooth estimating functions for both Gehan and Log-rank weights, reducing computational cost, and improving compatibility with standard numerical algorithms. Incorporating time-varying covariates, such as the Charlson comorbidity index, would provide a more dynamic view of patient health and better capture the evolving impact of treatment over time. To handle unobserved heterogeneity, a frailty component (Strawderman, 2006) could be integrated, supported by findings from Gonzalez et al. (2005), who reported lower readmission risks in women even after adjusting for clinical variables. Moreover, exploring nonlinear relationships using deep learning approaches—such as LSTM-Cox models (Runquan and Xiaoping, 2024)—could improve the modeling of temporal dependencies in recurrent event data. A promising direction also lies in adapting the deep neural network-based AFT models proposed by Kim et al. (2023),

which use Gehan rank loss for single failure time data, to censored recurrent event settings, potentially enhancing both flexibility and predictive accuracy.

APPENDIX A: REGULARITY CONDITIONS

The following regularity conditions ensure valid results and support asymptotic analysis. We assume that:

C1. $f_0(\cdot)$ satisfying $F_0(\cdot) = \int_0^{\cdot} f_0(s)ds$ has continuous first and bounded second derivatives
C2. The sequence of observable processes at time $s \in \mathcal{S}$: $\{(N_i^{\dagger}(s), Y_i^{\dagger}(s), \mathcal{E}_i(s), \mathbf{Z}_i), i = 1, \ldots, n\}$ are independent and identically distributed.
C3. $\tau_1 \mid \boldsymbol{Z}_1$ is a possibly degenerate distribution $G(. \mid \boldsymbol{Z}_1)$.
C4. The censorship is non-informative.
C5. $\boldsymbol{Z}_1$ is almost surely bounded by a nonrandom constant Γ.
C6. $\boldsymbol{z}^{(j)}(t \mid \boldsymbol{\theta})$, for $j = 0, 1, 2$, is continuous for $(t, \boldsymbol{\theta}) \in \mathbb{R}^+ \times \mathcal{N}(\boldsymbol{\theta}_0)$, where $\mathcal{N}(\boldsymbol{\theta}_0)$ is some neighbourhood of $\boldsymbol{\theta}_0$.
C7. Φ_G in (6.1) is a nonsingular matrix.
C8. $\bar{\tau}$ satisfies $\inf\left\{t \in [0, \bar{\tau}] : \boldsymbol{z}^{(0)}(t \mid \boldsymbol{\theta}_0)\right\} > 0$.
C9. For $(t, \boldsymbol{\theta}) \in [0, \bar{\tau}] \times \mathcal{N}(\boldsymbol{\theta}_0), \mathbb{W}(t \mid \boldsymbol{\theta})$ is of bounded variation and there exists a continuous, bounded deterministic function $w(t \mid \boldsymbol{\theta})$ such that
$|\mathbb{W}(t \mid \boldsymbol{\theta}) - w(t \mid \boldsymbol{\theta})|$ converges to zero uniformly in probability.
C10. Φ_W in (6.2) is a nonsingular.

APPENDIX B: PROOFS OF SECTION 4 PROPOSITIONS

PROOF OF PROPOSITION 4.1. The orthogonality condition (4.8) simplifies to:

$$\mathbb{E}\left[\left\{U_{\boldsymbol{\theta}_k}^{\psi}(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi_k}(\boldsymbol{\delta}_0)\right\} U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0)\right] = 0, \text{ for } k = 1, \ldots, p, \quad \xi_k(\cdot) \in \mathbb{H}, \tag{B.1}$$

and any function $\kappa(\cdot) \in \mathbb{R}$ not identically zero. Without loss of generality, we consider $k = 1$, as the argument holds for all $k = 1, \ldots, p$. To solve for $\xi_1(\cdot)$, we begin by rewriting the left-hand side of equation (B.1) using equations (4.2) and (4.4), which define $U_{\boldsymbol{\theta}_1}^{\psi}(\boldsymbol{\delta}_0)$ and $U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0)$, respectively. Since the $M_i^{\dagger}(\cdot)$ are orthogonal $\mathbf{F}$-martingales by Lemma 2.1, and under Condition C2, applying Proposition 3.1 yields:

$$LHS \equiv n \int_0^t \mathbb{E}\left[\left\{Z_{11}\psi(w) - \xi_1(w)\right\} \kappa(w)\lambda_0(w) Y_1(w \mid \boldsymbol{\theta}_0)\right] dw.$$

It follows that

$$\begin{aligned} (\text{B.1}) &\Longleftrightarrow \psi(w)\mathbb{E}\left[Z_{11}Y_1(w \mid \boldsymbol{\theta}_0)\right] - \xi_1(w)\mathbb{E}\left[Y_1(w \mid \boldsymbol{\theta}_0)\right] = 0 \\ &\Longleftrightarrow \xi_1(w) = \psi(w)\bar{Z}_1(w), \end{aligned}$$

with $\bar{Z}_1(w) = \mathbb{E}\left[Z_{11}Y_1(w \mid \boldsymbol{\theta}_0)\right] / \mathbb{E}\left[Y_1(w \mid \boldsymbol{\theta}_0)\right]$. The solution $\xi_1(\cdot)$ is an element of $\mathbb{H}$. In fact, by Condition C5, Z_1 is uniformly bounded. Furthermore, Condition A6 ensures that $\boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0) \neq 0$, which implies $\mathbb{E}\left[Y_1(r \mid \boldsymbol{\theta}_0)\right] = 0$ cannot be zero. Thus, using the score in (4.6) on the effective age scale, we obtain the efficient score function:

$$U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0) = \sum_{i=1}^{n} \int_0^t \psi(w)\left\{\boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(w)\right\} dM_i(w \mid \boldsymbol{\theta}_0).$$

Finally, the information bound $\mathcal{B}(\boldsymbol{\theta}_0)$ is the inverse of the efficient information matrix $\Sigma_{\boldsymbol{\theta}}^{\psi,\text{eff}}$, the covariance of $U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)$, computed via martingale properties:

$$\text{(B.2)} \qquad \begin{aligned}\Sigma_{\boldsymbol{\theta}}^{\psi,\text{eff}} =& n^{-1}\mathbb{E}\left[U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)\right] \\ =& \int_0^t \mathbb{E}\left[\left\{\boldsymbol{Z}_1 - \bar{\boldsymbol{Z}}(w)\right\}^{\otimes 2} Y_1\left(w \mid \boldsymbol{\theta}_0\right)\right]\psi^2(w)\lambda_0(w)dw,\end{aligned}$$

which ends the proof. □

PROOF OF PROPOSITION 4.2. The efficient score function for $\Lambda_0(\cdot)$ is defined as:

$$\text{(B.3)} \qquad U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0) = U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) - \Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0),$$

where $\Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) = \mathbb{E}\left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)\right]\left\{\mathbb{E}\left[U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)\right]\right\}^{-1} U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)$, is the projection of $U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)$ onto the space spanned by $U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)$. Substituting the formula for $\Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)$ into (B.3) and using (B.2), we obtain:

$$\text{(B.4)} \qquad U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0) = U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) - \frac{1}{n}\mathbb{E}\left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)\right]\mathcal{B}(\boldsymbol{\theta_0})U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0),$$

where $U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^\infty \frac{I\{\tilde{\mathcal{E}}_i(v)\le t\}}{\boldsymbol{z}^{(0)}\{\tilde{\mathcal{E}}_i(v)|\boldsymbol{\theta}_0\}} dM_i^{\dagger}(v)$, for $\nu(r) = I\left\{r \le t\right\}/\boldsymbol{z}^{(0)}(r \mid \boldsymbol{\theta}_0)$. Using Proposition 3.1, we show that $\mathbb{E}\left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)\right] = n\Delta_{\psi}(t)$, where $\Delta_{\psi}(t)$ is given in (4.12). This follows from the orthogonality of the $M_i^{\dagger}(\cdot)$ by Lemma 2.1, Condition C2, which ensures i.i.d. integrands, and subsequent use of Proposition 3.1, making the left-hand side:

$$\begin{aligned}LHS = \equiv n\mathbb{E}&\left[\int_0^\infty \left(\frac{I\left\{w \le t\right\}}{\boldsymbol{z}^{(0)}\left\{w \mid \boldsymbol{\theta}_0\right\}}\right)\boldsymbol{Z}_1\psi(w)\lambda_0(w)Y_1\left(w \mid \boldsymbol{\theta}_0\right)dw\right] \\ =& n\int_0^t \frac{\boldsymbol{z}^{(1)}\left\{w \mid \boldsymbol{\theta}_0\right\}}{\boldsymbol{z}^{(0)}\left\{w \mid \boldsymbol{\theta}_0\right\}}\psi(w)\lambda_0(w)dw.\end{aligned}$$

Note that $\Delta_{\psi}(t)$ and $U_{\boldsymbol{\theta}}^{\psi,\text{eff}}(\boldsymbol{\delta}_0)$ are p-column vectors while $\mathcal{B}(\boldsymbol{\theta_0})$ is a $p \times p$ square matrix. Consequently, $\Delta_{\psi}'(t)\mathcal{B}(\boldsymbol{\theta_0})\psi(w)\left\{\boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(w)\right\}$ is a scalar and (B.4) is well-defined, simplifying to:

$$U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^\infty \left[\frac{I\{w \le t\}}{\boldsymbol{z}^{(0)}\left(w \mid \boldsymbol{\theta}_0\right)} - \Delta_{\psi}'(w)\mathcal{B}(\boldsymbol{\theta_0})\psi(w)\left\{\boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(w)\right\}\right]dM_i(w \mid \boldsymbol{\theta}_0),$$

which gives (4.11). The information bound $\mathcal{B}(\Lambda_0)$ for estimating $\Lambda_0(t)$ requires computing $\text{Var}\left[n^{-1/2}U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)\right]$. Using the effective age representation of $U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)$, along with the variance formula in (3.7) and Condition C2, this variance is given by:

$$\int_0^\infty \mathbb{E}\left[\left(\frac{I\{w \le t\}}{\boldsymbol{z}^{(0)}\left(w \mid \boldsymbol{\theta}_0\right)} - \Delta_{\psi}'(w)\mathcal{B}(\boldsymbol{\theta_0})\psi(w)\left\{\boldsymbol{Z}_1 - \bar{\boldsymbol{Z}}(w)\right\}\right)^2\right]\lambda_0(w)Y_1\left(w \mid \boldsymbol{\theta}_0\right)dw.$$

Expanding and simplifying the squared term inside the expectation as follows:

$$\begin{aligned}\text{Var}\left[n^{-1/2}U_{\Lambda}^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)\right] =& \int_0^t \frac{\mathbb{E}[Y_1\left(w \mid \boldsymbol{\theta}_0\right)]\lambda_0(w)}{\boldsymbol{z}^{(0)}\left(w \mid \boldsymbol{\theta}_0\right)^2}dw - 2 \times \int_0^t \Delta_{\psi}'(w)\mathcal{B}(\boldsymbol{\theta_0})\psi(w) \\ &\times \left\{\frac{\mathbb{E}[\boldsymbol{Z}_1 Y_1\left(w \mid \boldsymbol{\theta}_0\right)]}{\boldsymbol{z}^{(0)}\left(w \mid \boldsymbol{\theta}_0\right)} - \bar{\boldsymbol{Z}}(w)\frac{\mathbb{E}[Y_1\left(w \mid \boldsymbol{\theta}_0\right)]}{\boldsymbol{z}^{(0)}\left(w \mid \boldsymbol{\theta}_0\right)}\right\}\lambda_0(w)dw\end{aligned}$$

$$+\int_0^t \mathbb{E}\left[\left[\Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta_0})\psi(w)\left\{\boldsymbol{Z}_1-\bar{\boldsymbol{Z}}(w)\right\}\right]^2 Y_1\left(w\mid\boldsymbol{\theta}_0\right)\right]\lambda_0(w)dw$$

$$=\int_0^t \frac{\lambda_0(w)}{s^{(0)}\left(w\mid\boldsymbol{\theta}_0\right)}dw+\Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta_0})\Delta_\psi(w)\,,$$

yields the information bound $\mathcal{B}(\Lambda_0)$ in (4.13). □

APPENDIX C: AUXILIARY RESULTS OF SECTION 5

We first introduce several auxiliary results, considering all probabilities and expectations with respect to $\mathbb{P}_0$, the probability measure corresponding to $\boldsymbol{\delta}_0$. Let Θ be an open convex subset of $\mathbb{R}^p$. The proofs of Theorems C.1 and C.2 are built upon the empirical process theory established in van der Vaart and Wellner (1996).

THEOREM C.1. *Let $\boldsymbol{\theta}\in\Theta$ be fixed, and for $t\in\mathbb{R}^+$, define*

$$\mathbb{N}^{(k)}(t\mid\boldsymbol{\theta})=\frac{1}{n}\sum_{i=1}^n \boldsymbol{Z}_i^{\otimes k}N_i(t\mid\boldsymbol{\theta})\quad(k=0,1,2). \tag{C.1}$$

When viewed as processes over $t\in\mathbb{R}^+$, both $\mathbb{N}^{(k)}(t\mid\boldsymbol{\theta})$ in (C.1) and

$$\mathbb{Z}^{(k)}(t\mid\boldsymbol{\theta})=\frac{1}{n}\sum_{i=1}^n \boldsymbol{Z}_i^{\otimes k}Y_i(t\mid\boldsymbol{\theta}),\quad(k=0,1,2), \tag{C.2}$$

are $\mathbb{P}_0$-Donsker.

THEOREM C.2. *Both (C.1) and (C.2) are $\mathbb{P}_0$-Donsker as processes indexed by $(t,\boldsymbol{\theta})\in\mathbb{R}^+\times\Theta$.*

The corollary below follows directly from Theorems C.1 and C.2.

COROLLARY C.1. *Under Conditions C1–C5, for $k=0,1,2$, the following convergence in probability holds:*

$$\sup_{(t,\boldsymbol{\theta})\in\mathbb{R}^+\times\Theta}\left\|\mathbb{N}^{(k)}(t\mid\boldsymbol{\theta})-\boldsymbol{n}^{(k)}(t\mid\boldsymbol{\theta})\right\|\to 0,\qquad \sup_{(t,\boldsymbol{\theta})\in\mathbb{R}^+\times\Theta}\left\|\mathbb{Z}^{(k)}(t\mid\boldsymbol{\theta})-\boldsymbol{z}^{(k)}(t\mid\boldsymbol{\theta})\right\|\to 0,$$

where $\boldsymbol{n}^{(k)}(t\mid\boldsymbol{\theta})=\mathbb{E}\left[\boldsymbol{Z}_1^{\otimes k}N_1(t\mid\boldsymbol{\theta})\right]$. These results remain valid if the supremum is taken only over $t\in\mathbb{R}^+$ while keeping $\boldsymbol{\theta}\in\Theta$ fixed.

We now give a useful approach for evaluating $\boldsymbol{n}^{(k)}(t\mid\boldsymbol{\theta})$ and $\boldsymbol{z}^{(k)}(t\mid\boldsymbol{\theta})$, for $k=0,1,2$.

LEMMA C.1. *Let $\boldsymbol{\theta}\in\mathbb{R}^p$ be arbitrary. Then, for each $i=1,\ldots,n$, the counting and at-risk processes satisfy the transformations:*

$$N_i(t\mid\boldsymbol{\theta})=N_i\left(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i}\mid\boldsymbol{\theta_0}\right),\quad Y_i(t\mid\boldsymbol{\theta})=Y_i\left(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i}\mid\boldsymbol{\theta_0}\right). \tag{C.3}$$

Furthermore, for $k=0,1,2$, the expectation of the weighted counting process can be rewritten as:

$$\boldsymbol{n}^{(k)}(t\mid\boldsymbol{\theta})=\mathbb{E}\left[\boldsymbol{Z}_1^{\otimes k}\int_0^{te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_1}}Y_1(w\mid\boldsymbol{\theta}_0)\lambda_0(w)dw\right]. \tag{C.4}$$

PROOF. We prove the results in (C.3) step by step. Starting from (3.2), we have:

$$N_i(t \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} I(\tilde{\mathcal{E}}_i(S_{i,j}) \leq te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i}) = N_i(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \mid \boldsymbol{\theta}_0).$$

Next, from (3.4), as $s \to \infty$, we have:

$$Y_i(t \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} \frac{I_{\mathcal{E}_{i,j-1}}(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \cdot e^{-\boldsymbol{\theta}_0'\boldsymbol{Z}_i}; S_{i,j-1}, S_{i,j})}{\mathcal{E}'_{i,j-1}(\mathcal{E}^{-1}_{i,j-1}(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \cdot e^{-\boldsymbol{\theta}_0'\boldsymbol{Z}_i}))} + \frac{I_{\mathcal{E}_{i,K_i}}(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \cdot e^{-\boldsymbol{\theta}_0'\boldsymbol{Z}_i}; S_{i,K_i}, \tau_i)}{\mathcal{E}'_{i,K_i}(\mathcal{E}^{-1}_{i,K_i}(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \cdot e^{-\boldsymbol{\theta}_0'\boldsymbol{Z}_i}))}$$
$$= Y_i(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_i} \mid \boldsymbol{\theta}_0).$$

Finally, to prove (C.4), note that the process $\{M_i(t \mid \boldsymbol{\theta}_0) : t \in \mathcal{T}\}$ is not a martingale but has zero mean. Therefore, for $k = 0, 1, 2$,

$$\boldsymbol{n}^{(k)}(t \mid \boldsymbol{\theta}) = \mathbb{E}\left[\boldsymbol{Z}_1^{\otimes k}\left\{M_1(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_1} \mid \boldsymbol{\theta}_0) + A_1(te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_1} \mid \boldsymbol{\theta}_0)\right\}\right]$$
$$= \mathbb{E}\left[\boldsymbol{Z}_1^{\otimes k}\int_0^{te^{(\boldsymbol{\theta}_0-\boldsymbol{\theta})'\boldsymbol{Z}_1}} Y_1(w \mid \boldsymbol{\theta}_0)\lambda_0(w)dw\right],$$

which ends the proof. □

APPENDIX D: PROOFS OF SECTION 5 THEOREMS

PROOF OF THEOREM 6.1. The efficient score function for $\boldsymbol{\theta}$ can be rewritten as:

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \int_0^\infty \mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta})d\mathbb{N}^{(1)}(w \mid \boldsymbol{\theta}) - \int_0^\infty \mathbb{Z}^{(1)}(w \mid \boldsymbol{\theta})d\mathbb{N}^{(0)}(w \mid \boldsymbol{\theta}).$$

First, by applying Theorem C.1 and Corollary C.1, we can use Lemma 1 from Jin et al. (2006). Consequently, there exists a function ϕ such that:

$$n^{-\frac{1}{2}}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) = n^{-\frac{1}{2}}\sum_{i=1}^n \phi_i + o_p(1),$$

where $\phi_i = \int_0^\infty \left\{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\boldsymbol{Z}_i - \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right\} dM_i(w \mid \boldsymbol{\theta}_0)$, and satisfies the condition:

$$\mathbb{E}[\phi_1] = \mathbb{E}\left[\int_0^\infty \left\{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\boldsymbol{Z}_1 - \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right\} dM_1(w \mid \boldsymbol{\theta}_0)\right] = 0,$$

by (3.6). Using Lemma 3.1. Also, from (3.7), we have:

$$Var(\phi_1) = \int_0^\infty \mathbb{E}\left[\left\{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\boldsymbol{Z}_1 - \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right\}^{\otimes 2} Y_1(w \mid \boldsymbol{\theta}_0)\right]\lambda_0(w)dw$$
$$= \int_0^\infty \left[\left\{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\right\}^2 \boldsymbol{z}^{(2)}(w \mid \boldsymbol{\theta}_0) - \boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\left\{\boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0)\right\}^{\otimes 2}\lambda_0(w)\right]dw,$$

which shows that $Var(\phi_1) = \Sigma_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}$. The central limit theorem implies that $n^{-\frac{1}{2}}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) \xrightarrow{d} N\left(0, \Sigma_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}\right)$, with Conditions C1 and C6 ensuring that $\left\|\Sigma_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}\right\| < \infty$. Next, under Conditions C1–C6, along with Corollary C.1 and Lemma C.1, the function $n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ converges

uniformly in probability to:

$$\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \int_0^\infty \Big\{ \boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}) \mathbb{E}\left[\boldsymbol{Z}_1 Y_1(w \mid \boldsymbol{\theta}) \lambda_0 \left\{ w c_1(\boldsymbol{\theta}) \right\} c_1(\boldsymbol{\theta})\right] - \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}) \mathbb{E}\left[Y_1(w \mid \boldsymbol{\theta}) \lambda_0 \left\{ w c_1(\boldsymbol{\theta}) \right\} c_1(\boldsymbol{\theta})\right] \Big\} dw,$$

with $c_1(\boldsymbol{\theta}) = \exp\left\{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \boldsymbol{Z}_1\right\}$ and Conditions C6 and C7 ensuring that $\boldsymbol{\theta}_0$ is unique. Moreover, given that $c_1(\boldsymbol{\theta}_0) = 1$, $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0)$ satisfies the condition:

$$\begin{aligned}\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) &= \int_0^\infty \Big\{ \boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0) \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0) - \boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta}_0) \boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0) \Big\} \lambda_0(w) dw \\ &= 0.\end{aligned}$$

Hence, under Conditions C1–C6, the function $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0)$ has a non-singular derivative matrix, given by $\Phi_G = \dot{\mathscr{U}}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0)$, which is continuous at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Moreover, both $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ and $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ exhibit component-wise monotonicity in $\boldsymbol{\theta}$. Finally, let $C_{\epsilon_n} = \{a \in \mathbb{R}^p : \|a\| \leq \epsilon_n\}$ denote a sequence of compact sets, indexed by an arbitrary constant $0 < \epsilon_n < \infty$. To complete the argument, we must show that the following holds:

$$\text{(D.1)} \qquad \sup_{(\boldsymbol{\theta}-\boldsymbol{\theta}_0) \in C_{\epsilon_n}} n^{\frac{1}{2}} \left\{ \frac{n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) - n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) - \mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})}{1 + n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|} \right\} \xrightarrow{P} 0,$$

for any positive sequence $\epsilon_n \to 0$. First, we define $\bar{\delta}_U = n^{-1}\left\{U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) - U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0)\right\}$. Then we derive a closed-form expression for $\bar{\delta}_U$ as:

$$\begin{aligned}\bar{\delta}_U = &\int_0^\infty \mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta}) d\mathbb{N}^{(1)}(w \mid \boldsymbol{\theta}) - \int_0^\infty \mathbb{Z}^{(1)}(w \mid \boldsymbol{\theta}) d\mathbb{N}^{(0)}(w \mid \boldsymbol{\theta}) \\ &- \int_0^\infty \mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta}_0) d\mathbb{N}^{(1)}(w \mid \boldsymbol{\theta}_0) + \int_0^\infty \mathbb{Z}^{(1)}(w \mid \boldsymbol{\theta}_0) d\mathbb{N}^{(0)}(w \mid \boldsymbol{\theta}_0).\end{aligned}$$

By performing a change of variable in the last two terms of the expansion, applying Lemma C.1, and factoring out the common terms, we obtain:

$$\bar{\delta}_U = n^{-1} \sum_{i=1}^n \int_0^\infty \left\{\mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta}) \boldsymbol{Z}_i - \mathbb{Z}^{(1)}(w \mid \boldsymbol{\theta})\right\} Y_i(w \mid \boldsymbol{\theta}) \times \left[\lambda_0 \left\{ w c_i(\boldsymbol{\theta}) \right\} c_i(\boldsymbol{\theta}) - \lambda_0(w)\right] dw.$$

Now, we consider a positive sequence of $\epsilon_n \to 0$, such that $\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \leq \epsilon_n$. Then using Condition C1, a Taylor series expansion of $\lambda_0 \left\{ r c_1(\boldsymbol{\theta}) \right\} c_1(\boldsymbol{\theta})$ in the neighborhood of $\boldsymbol{\theta}_0$ yields:

$$\lambda_0 \left\{ r c_1(\boldsymbol{\theta}) \right\} c_1(\boldsymbol{\theta}) = \lambda_0(r) - l_0(r) \boldsymbol{Z}_1' (\boldsymbol{\theta} - \boldsymbol{\theta}_0) + o_P\left(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2\right),$$

where $l_0(r) = \lambda_0(r) + r\dot{\lambda}_0(r) = \psi(r)\lambda_0(r)$. Substituting these into $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$, we have:

$$\begin{aligned}\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = &- (\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \int_0^\infty \left[\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}) \boldsymbol{z}^{(2)}(w \mid \boldsymbol{\theta}) - \left\{\boldsymbol{z}^{(1)}(w \mid \boldsymbol{\theta})\right\}^{\otimes 2}\right] \psi(w) \lambda_0(w) dw \\ &+ o_P\left(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|\right).\end{aligned}$$

Φ_G in (6.1) is obtained by taking the derivative of $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ with respect to $\boldsymbol{\theta}$ in the latter equation. Similarly, we substitute the Taylor series expansion of $\lambda_0\left\{rc_1(\boldsymbol{\theta})\right\}c_1(\boldsymbol{\theta})$ into $\bar{\delta}_U$ to obtain:

$$\bar{\delta}_U = -\left(\boldsymbol{\theta}-\boldsymbol{\theta}_0\right)'\int_0^{\infty}\left[\mathbb{Z}^{(0)}(w\mid\boldsymbol{\theta})\mathbb{Z}^{(2)}(w\mid\boldsymbol{\theta})-\left\{\mathbb{Z}^{(1)}(w\mid\boldsymbol{\theta})\right\}^{\otimes 2}\right]\psi(w)\lambda_0(w)dw + o_P\left(\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|\right).$$

Then subtracting $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ from $\bar{\delta}_U$ results in:

$$\bar{\delta}_U - \mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = -\left(\boldsymbol{\theta}-\boldsymbol{\theta}_0\right)'\int_0^{\infty}\Bigg[\left\{\mathbb{Z}^{(0)}(w\mid\boldsymbol{\theta})\mathbb{Z}^{(2)}(w\mid\boldsymbol{\theta})-\boldsymbol{z}^{(0)}(w\mid\boldsymbol{\theta})\boldsymbol{z}^{(2)}(w\mid\boldsymbol{\theta})\right\} - \left\{\left\{\mathbb{Z}^{(1)}(w\mid\boldsymbol{\theta})\right\}^{\otimes 2}-\left\{\boldsymbol{z}^{(1)}(w\mid\boldsymbol{\theta})\right\}^{\otimes 2}\right\}\Bigg]\psi(w)\lambda_0(w)dw.$$

This implies that:

$$\left\|n^{\frac{1}{2}}\frac{n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})-n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0)-\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})}{1+n^{\frac{1}{2}}\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|}\right\| \leq \frac{n^{\frac{1}{2}}\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|\,\|\mathbb{A}\|}{1+n^{\frac{1}{2}}\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|}. \tag{D.2}$$

Hence, by Theorem C.2, $\|\mathbb{A}\| \xrightarrow{P} 0$ and given that $\epsilon_n \to 0$, we obtain (D.3), which ends the proof. □

PROOF OF THEOREM 6.2. The efficient score function for $\boldsymbol{\theta}$ under a general weight function $\mathbb{W}(t\mid\boldsymbol{\theta})$ is given by:

$$U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^{n}\int_0^t \mathbb{W}(w\mid\boldsymbol{\theta})\left[\boldsymbol{Z}_i-\bar{\mathbb{Z}}(w\mid\boldsymbol{\theta})\right]dN_i(w\mid\boldsymbol{\theta}).$$

First, by applying Theorem C.1 and Corollary C.1, Lemma 1 from Jin et al. (2006) implies that there exists a function φ such that:

$$n^{-\frac{1}{2}}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0) = n^{-\frac{1}{2}}\sum_{i=1}^{n}\varphi_i + o_p(1),$$

where $\varphi_i = \int_0^{\bar{\tau}} w\left(r\mid\boldsymbol{\theta}_0\right)\left\{\boldsymbol{Z}_i-\bar{\boldsymbol{Z}}(r)\right\}dM_i\left(r\mid\boldsymbol{\theta}_0\right)$, and satisfies the condition:

$$\mathbb{E}[\varphi_1] = \mathbb{E}\left[\int_0^{\bar{\tau}} w\left(r\mid\boldsymbol{\theta}_0\right)\left\{\boldsymbol{Z}_i-\bar{\boldsymbol{Z}}(r)\right\}dM_1\left(r\mid\boldsymbol{\theta}_0\right)\right] = 0,$$

by (3.6). Using Lemma 3.1, we have from (3.7):

$$\begin{aligned} Var\left(\varphi_1\right) &= \int_0^{\bar{\tau}} w^2\left(r\mid\boldsymbol{\theta}_0\right)\mathbb{E}\left[\left\{\boldsymbol{Z}_i-\left\{\boldsymbol{z}^{(1)}\left(r\mid\boldsymbol{\theta}_0\right)/\boldsymbol{z}^{(0)}\left(r\mid\boldsymbol{\theta}_0\right)\right\}\right\}^{\otimes 2}Y_1\left(r\mid\boldsymbol{\theta}_0\right)\right]\lambda_0(r)dr \\ &= \int_0^{\bar{\tau}} w^2\left(r\mid\boldsymbol{\theta}_0\right)\left[\frac{\boldsymbol{z}^{(2)}\left(r\mid\boldsymbol{\theta}_0\right)}{\boldsymbol{z}^{(0)}\left(r\mid\boldsymbol{\theta}_0\right)}-\left\{\frac{\boldsymbol{z}^{(1)}\left(r\mid\boldsymbol{\theta}_0\right)}{\boldsymbol{z}^{(0)}\left(r\mid\boldsymbol{\theta}_0\right)}\right\}^{\otimes 2}\right]\boldsymbol{z}^{(0)}(r\mid\boldsymbol{\theta}_0)\lambda_0(r)dr, \end{aligned}$$

which shows that $Var\left(\varphi_1\right) = \Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}$. Consequently, the central limit theorem implies that $n^{-\frac{1}{2}}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0) \xrightarrow{d} N\left(0,\Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\right)$, with Condition C9 ensuring that $\left\|\Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}\right\| < \infty$. Under Conditions C1–C9, along with Corollary C.1 and Lemma C.1, the function $n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta})$ converges uniformly in probability to a function $\mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta})$, for $\boldsymbol{\theta}\in\mathscr{N}(\boldsymbol{\theta}_0)$. Conditions C6

and C10 guarantee the uniqueness of $\boldsymbol{\theta}_0$. Thus, we define $\Phi_W = \dot{\mathscr{U}}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0)$, whose existence is ensured by Conditions A1–A9, while Condition C10 guarantees its nonsingularity. Finally, let $C_{\epsilon_n} = \{a \in \mathbb{R}^p : \|a\| \leq \epsilon_n\}$ represent a sequence of sets, which may not necessarily be compact, indexed by an arbitrary constant $0 < \epsilon_n < \infty$. To conclude the proof, it remains to establish the following:

$$\text{(D.3)} \qquad \sup_{(\boldsymbol{\theta}-\boldsymbol{\theta}_0)\in C_{\epsilon_n}} n^{\frac{1}{2}} \left\{ \frac{n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}) - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0) - \mathscr{U}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta})}{1 + n^{\frac{1}{2}}\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|} \right\} \xrightarrow{P} 0,$$

for any positive sequence $\epsilon_n \to 0$. This condition follows from the same reasoning as in the proof of Theorem 6.1. Therefore, applying Theorem 7.3.2 of Bickel et al. (1993), the results immediately follow. □

PROOF OF THEOREM 6.3. Letting $n \to \infty$, we obtain the following limits:

$$\lim_{B\to\infty} \frac{1}{B}\mathfrak{P}'\mathfrak{P} = \begin{bmatrix} 1 & 0' \\ 0 & n^{-1}\widehat{\mathbb{A}}_G \end{bmatrix}, \lim_{B\to\infty} \frac{1}{B}\mathfrak{P}'\mathfrak{Q} = \begin{bmatrix} 0' \\ n^{-1}\widehat{\mathbb{A}}_G\Phi_W \end{bmatrix} + o_P(n^{-1}),$$

almost surely, given the data. It follows that $\widehat{\Phi}_W = \Phi_W + o_P(1)$, and thus the second result follows from this convergence along with the fact that $\widehat{\Sigma}_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}} \xrightarrow{P} \Sigma_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}$ under Conditions C1–C10. □

PROOF OF THEOREM 6.4. We can write $\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}) - \Lambda_0(t) = \delta_1 + \delta_2$, with $\delta_1 = \left\{\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}) - \widehat{\Lambda}_0(t \mid \boldsymbol{\theta}_0)\right\}$ and $\delta_2 = \left\{\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}_0) - \Lambda_0(t)\right\}$. We consider first δ_1. Under Conditions C1–C10, we can apply Corollary C.1 along with a reasoning similar to that in the proofs of Theorems 6.1 and 6.2 to prove the existence of a function χ such that:

$$\delta_1 = n^{-1}\sum_{i=1}^{n} \chi_i + o_P\left(n^{-\frac{1}{2}}\right),$$

where $\chi_i = \int_0^t \frac{Y_i(w|\boldsymbol{\theta})}{\boldsymbol{z}^{(0)}(w|\boldsymbol{\theta})}\left[\lambda_0\left\{wc_i(\boldsymbol{\theta})\right\}c_i(\boldsymbol{\theta}) - \lambda_0(w)\right]dw$. Then, assuming that $\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\| \leq \epsilon_n$ for a positive sequence $\epsilon_n = n^{-\frac{1}{2}}\Gamma \to 0$, we have $\delta_1 = -\Delta'_\psi(t)\left(\boldsymbol{\theta}-\boldsymbol{\theta}_0\right) + o_P\left(n^{-\frac{1}{2}}\right)$, where $\Delta_\psi(t)$ is given in (4.12). This yields the second term of (6.5), showing that $\widehat{\Lambda}_0(t \mid \boldsymbol{\theta})$ is asymptotically linear in $\boldsymbol{\theta}$. Considering now δ_2, we can use the same Corollary C.1 to show that there exist a function γ, such that:

$$\delta_2 = n^{-1}\sum_{i=1}^{n} \gamma_i + o_p\left(n^{-\frac{1}{2}}\right),$$

with $\gamma_i = \int_0^t \left\{\boldsymbol{z}^{(0)}(w \mid \boldsymbol{\theta}_0)\right\}^{-1} dM_i\left(w \mid \boldsymbol{\theta}_0\right)$. Substituting $\widehat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0 = -n^{-1}\Phi_W^{-1}U_{\boldsymbol{\theta}}^{\mathbb{W},\text{eff}}(\boldsymbol{\delta}_0) + o_P(n^{-\frac{1}{2}})$ into δ_1 and by (6.5), $\widehat{\Lambda}_0(t \mid \boldsymbol{\theta}) - \Lambda_0(t)$ converges uniformly in probability to $\bar{\varrho}$, where:

$$\varrho_i\left(t \mid \boldsymbol{\theta}_0\right) = \Delta'_\psi(t)\Phi_W^{-1}\int_0^{\bar{\tau}} w\left(r \mid \boldsymbol{\theta}_0\right)\left\{\boldsymbol{Z}_i - \bar{\boldsymbol{Z}}(r)\right\} dM_i\left(r \mid \boldsymbol{\theta}_0\right) + \gamma_i$$

Finally, the function $\varrho(\cdot \mid \boldsymbol{\theta}_0)$ is càdlàg and has bounded variation for $t \in [0, \bar{\tau}]$. As a result, it is $\mathbb{P}_0$-Donsker. Hence, according to Parner (1998), the functional central limit theorem for uniformly bounded empirical processes ensures that the sequence $n^{-\frac{1}{2}}\sum_{i=1}^{n}\gamma_i$ converges in distribution to a Gaussian process. Its covariance is given by $\mathbb{E}\left\{\varrho_1(r \mid \boldsymbol{\theta}_0)\varrho_1(t \mid \boldsymbol{\theta}_0)\right\}$, for any $r, t \in [0, \tau]$. A direct calculation using Lemma 3.1 and arguments from Proposition 4.2 confirms the covariance expression (6.4); details are omitted for brevity. □

REFERENCES

[1] BILLINGSLEY, P. (1999). *Convergence of Probability Measures*, 2nd ed. Wiley, New York.

[2] BOURBAKI, N. (1966). *General Topology* **1**. Addison–Wesley, Reading, MA.

[3] ETHIER, S. N. and KURTZ, T. G. (1985). *Markov Processes: Characterization and Convergence*. Wiley, New York.

[4] PROKHOROV, YU. (1956). Convergence of random processes and limit theorems in probability theory. *Theory Probab. Appl.* **1** 157–214.